Core Technology Overview

Detailed Description · v13 (Pass 118: §109.9 Contextual Value Models; Pass 117 §1.5 The Fact Unit Language [claim 342]; §1.4 Record-Fused Encoding) · 293 registry features · 354 claims · Appendix A carries 293 features · Confidential · v14 addendum, October 8, 2026: Sections 50–55 and implementation updates (prototype v9.64) · v15 addendum, October 22, 2026: Section 56 The Self-Knowledge Layer · Registry v70: 391 features · 362 claims

1. The Fact Unit

The fundamental data primitive of the QueryBook system is the Fact Unit: a discrete, atomic, provenance-tracked knowledge record, identified by a Fact Unit Identifier (FUID) deterministically derived from the record's content, source, and ingestion timestamp. Each Fact Unit carries a confidence value, a safety classification, a language-independent semantic fingerprint, an embedding representation, temporal anchoring, and a reference into a provenance chain, and is immutable once created — a revision produces a new, linked version rather than modifying the existing record.

The confidence value is not a scalar. It is maintained as a first parameter accumulating evidence supporting the assertion and a second accumulating evidence contradicting it, the reported confidence being the ratio of the first to their sum, and their sum constituting the evidential weight from which a credible interval and a propagation precision are computed. The contributed weight is further attributed across registered source classes, from which a diversity measure is computed. Two records reporting an identical confidence and resting on different quantities or varieties of evidence are therefore distinguishable, which a scalar could not express.

1.1 Precise Definition

A Fact Unit is a record encoding exactly one semantic assertion, comprising the following fields, of which those marked required are present in every Fact Unit without exception and a candidate lacking any of them is refused at construction rather than admitted and flagged.

Field	Required	Content
FUID	Required	An identifier derived deterministically from the record's content, its source and its ingestion time. Identical content from an identical source at an identical time yields an identical identifier.
Type reference	Required	A reference into the governed

Field	Required	Content
		event and entity type catalogue, against whose machine definition the record is validated.
Subject reference	Required	A canonical, language-neutral identifier within the concept structure.
Predicate reference	Required	A canonical identifier drawn from the governed predicate catalogue. A predicate not in the catalogue cannot be asserted.
Object	Required	An entity reference or a typed scalar carrying its unit and dimension.
Temporal anchor	Required	When the assertion held, distinguished from when it was ingested and from when its source was created.
Spatial anchor	Optional	Position, extent or containment, where the assertion carries one.
Confidence value	Required	A pair of parameters: the first accumulating evidence supporting the assertion, the second accumulating evidence contradicting it. The reported confidence is the ratio of the first to their sum; the sum is the evidential weight, from which a credible interval and a propagation precision are computed.
Source-class attribution	Required	The distribution of contributed weight across registered source classes, from which the diversity measure is computed.
Provenance reference	Required	A reference into the append-only provenance chain, whose nodes carry keyed checksums and a deployment signature.

Field	Required	Content
Semantic fingerprint	Required	A language-independent value over the canonicalised triple. Two records asserting the same thing in different languages carry the same fingerprint and converge upon one record with two provenance branches.
Modality indication	Required	The modality from which the assertion was extracted.
Safety classification	Required	Carried within the record rather than applied around it, such that no path reaches the assertion without it.
Access control list	Required	Likewise carried within the record. Entitlement is a property of the assertion, not a filter over a result.
Typed edges	Optional	Weighted relations to other records, and membership in hyperedges relating sets.
Supersession chain	Optional	A link to the version this record replaces, and the relation — correction or restatement — under which it replaces it.
Derivation marking	Conditional	Present where the record was derived rather than reported: an inference marked derived, a reconstruction marked reconstructed, or an assertion generated by a named external engine. Unmodifiable and unremovable.

Three properties follow from the schema and are stated because each is relied upon elsewhere. A Fact Unit is immutable: a revision produces a new linked version and never modifies an existing record, so a provenance chain is append-only in fact and not merely by policy. Safety classification and entitlement are fields rather than layers, so no code path reaches an assertion without them and rights enforcement is structural rather than best-effort. And the assertion is one

assertion: a passage stating three things becomes three records, which is what permits a response to cite the specific claim it relied upon rather than the paragraph containing it.

1.2 The Post-Baconian Layered Definition

The precise definition of §1.1 remains in force whole; this subsection organizes it and states the enrichment the Baconian feedback loop provides. The Fact Unit reads as nine layers: a core semantic identity that never changes (FUID, type reference, canonical subject, predicate, and object references, and the semantic fingerprint — an assertion, not a concept; concepts belong to the World-View Model); structural adjacency (typed weighted edges and hyperedge memberships in the semantic, temporal, editorial, media, and language classes); temporal stability, read from the three timestamps and the supersession chain rather than asserted; edition-differential markers carried by edition version chains; multi-language alignment anchored to the fingerprint, with wording-operative spans reproduced verbatim; media provenance, with cross-modality corroboration and introduced content marked and inadmissible as grounding; the observation-and-evidence layer — the Baconian core — where the evidence pair is the observation count, diversity is the effective number of contributing source classes, contradictions are first-class and retained even when too weak to revise, and every update is closed-form and free of sampling; in-record rights and safety; and context-keyed deterministic output. The Baconian feedback loop — action, observation, measurement, adjustment, iteration — drives the enrichment: each observation increments the evidence pair by its source-reliability weight, diversity gates generalization, revision is precision-weighted so a settled assertion resists by arithmetic, and adjustment reaches only the layers the loop owns, never the semantic core. Specified at §3.6 and §3.7.

1.3 The Enhanced Record

Seven additional field families enrich the record, each an additive extension of the definitions above: no field of §1.1 and no layer of §1.2 is removed or altered, concepts continue to belong to the World-View Model, and every new field is a deterministic function of record content and the observation stream, so the determinism guarantee is preserved without qualification. First, uncertainty decomposition: residual uncertainty on the evidence pair is classified deterministically as epistemic — reducible by further observation — or aleatory — the world's own irreducible variability — with the source-reliability weighting, measurement precision, and consensus breadth carried as explicit sub-parameters; the classification gives the Baconian feedback loop a principled stopping condition, iterating

while uncertainty is epistemic and standing down when only the world's variance remains. Second, assertion polarity: every record is explicitly affirmative or negative, so a negative finding is a first-class record, contradiction detection gains a structural fast path — same assertion core, opposite polarity — and the Without-Information Rule distinguishes the absence of any record, which obliges refusal, from a known negative, which is an answer. Third, native n-ary assertion structure: one record carries one assertion with multiple typed argument slots — subject, value, unit, instrument, method, condition, observer — with one FUID, one fingerprint computed over the complete argument structure, and one evidence pair, rather than a constellation of binary statements joined by synthetic nodes. Fourth, applicability scope: the record states the conditions under which its assertion holds — temporal interval, jurisdiction, edition, parameter range, population — the record-side complement of the context lock, disciplining both retrieval and contradiction detection, since assertions under disjoint scopes do not conflict. Fifth, the Baconian adjustment history: every pass of the feedback loop appends an immutable entry — observation, source class, weighting applied, movement of the evidence pair and credible interval — so a record's confidence is replayable from first observation to present state, deterministically and exactly. Sixth, a cryptographic integrity chain: each record hashes its immutable core and each supersession link carries the hash of the record it supersedes, so version chains are tamper-evident by standard hashing alone — integrity machinery internal to the Fact Graph, with no distributed ledger, no token, and no ownership identity node. Seventh, a canonical compact encoding: a lossless, bidirectional binary serialization with required fields at fixed positions and optional fields tagged, under which FUID and fingerprint are identical in either form, making micro-embedded and offline deployment concrete and giving semantic media compression a specified record-level mechanism. These fields are consumed across the architecture: precision-weighted error propagation reads the decomposed uncertainty, contradiction detection and Competitive Candidate Resolution read polarity and scope, ingestion computes fingerprints over n-ary structure, supersession maintains the integrity chain, and deployment serializes to canonical form. Specified at §3.8.

1.4 Record-Fused Encoding and Database Neutrality

The canonical compact encoding of §1.3 is the Fact Unit's own at-rest and wire form, fused to the record rather than to any store, and this fusion is what lets the system operate over any database while carrying its guarantees across the storage boundary. A record is carried as itself — required fields at fixed offsets, optional fields tagged, FUID and semantic

fingerprint identical to the full form — so the encoded record is self-describing without an external schema. A host database, content repository, or e-reader platform holds the encoded record as an opaque value — a blob in a column, a file, an object, or a key's payload — and is never required to parse it, index its interior, or understand its fields; its only obligations are to accept bytes and return them unchanged. This is the invariant that preserves database neutrality: because the store is asked only to hold and return bytes, any database qualifies, and no proprietary format, schema migration, or query-engine capability is imposed on it, while the record's own FUID and fingerprint are exposed as plain opaque keys where an index is wanted, so retrieval by identity needs no reading of the record's meaning. Queries over Fact Units are expressed internally in terms of the record — its identifiers, fingerprint, typed adjacency, confidence, and applicability scope — and resolved against the Fact Graph by the system's own retrieval; where raw material must be drawn from an external store, an adapter translates the request into that store's native language (a relational query, an object lookup, a key fetch), and the external store never receives the internal fact-query form. An external store is thereby a source and a repository, never the reasoning engine, which is the position that keeps it replaceable and the determinism and groundedness guarantees independent of it. Because the store holds the record as an opaque blob, the properties the record carries travel with it: the integrity hash chain makes any alteration detectable on read, the in-record safety classification and access-control list keep entitlement a property of the record rather than a filter the store is trusted to apply, and under the zero-knowledge and key-custody provisions an encoded record written to an external store is unreadable by that store. Database neutrality is therefore not merely preserved but strengthened — any store will hold the record, and no store can read, reorder, or leak its meaning. The design invariant is stated plainly: the encoding and the query interface are fused to the Fact Unit and never to a host system's storage or query interface, so no host store is ever required to understand the record's format or speak its query language in order to store or return it. Specified at §3.9.

1.5 The Fact Unit Language

Fact Units are written in the Fact Unit Language: a formal, language-independent assertion notation in which every record is written and every mechanism reads. The notation is not a new field or a store — it names, and specifies as a language, what the record already is. Its vocabulary is the governed predicate catalogue; its grammar is the record schema of §1.1; and its identity function is the language-independent semantic

fingerprint, under which two statements asserting the same proposition in any two natural languages are the same statement. A statement comprises the canonical subject, predicate, and object identifiers drawn from the governed catalogues, a typed object value carrying its unit and dimension where the object is a scalar, the temporal anchor, and the record's evidence, provenance, safety, and entitlement fields; a candidate that does not conform is refused at admission rather than admitted incomplete. Because the language is formal, a Fact Unit can be composed, validated against the governed type catalogue, compared for contradiction, evaluated deterministically, and regenerated into any natural language; because it sits below language, nothing written in it is entangled with the words of its source, which is what makes one knowledge substrate serve every language and every modality at once, and it is why retrieval, contradiction detection, and regeneration each operate upon the assertion language rather than upon any natural-language text. Because the claim recites the record as a language rather than as a field list, a competitor offering the same benefits by a differently-fielded record still operates upon the same assertion language — which is why this is the element-light claim hardest to design around. Specified at §3.10 and recited as provisional claim 342.

2. What QueryBook Is and Is Not

QueryBook is not a LLM, Agent, RAG system, ChatBot or a basic Q&A system based on a LLM.

QueryBook is a fact-grounded knowledge system, embodied primarily as a question-answering system and secondarily as a multi-mode interactive platform. Every system output is derived from Fact Units — discrete, typed, atomic knowledge records, each carrying a confidence score, a safety classification, and a cryptographically verifiable provenance chain — rather than generated as unconstrained text.

The system further comprises a Neuromorphic Cognitive Layer implementing retrieval-stabilizing convergence dynamics, temporal sequence prediction, and anticipatory pre-activation, unified with a broader Predictive Cognitive Architecture (PCA) that organizes eight independently-arising prediction mechanisms — including those of the Neuromorphic Cognitive Layer itself — together with four additional mechanisms that complete a structural correspondence to an established two-part neuroscience framework, Predictive Coding and the Thousand Brains Theory of Intelligence, and three amplifications extending the resulting architecture's predictive reach across sessions and over time. This architecture predicts at the level of semantic meaning rather than

linguistic surface form, distinguishing it structurally from token-prediction-based language models.

The system provides two distinct, non-overlapping correctness guarantees: deterministic execution for a bounded, non-Turing-complete program representation compiled from natural-language instructions, and structural groundedness of generated natural-language output, in which the permissible output vocabulary is constrained to terms traceable to specific underlying Fact Units. An optional formal-verification subsystem applies machine-checked proof to precisely specifiable properties of the bounded program representation, presenting the resulting proof as evidence to a human reviewer who retains sole authority to approve any resulting change; a fixed, permanently-excluded set of subsystems is structurally unreachable by any proposal regardless of the evidence offered in its support.

The system further comprises an immersive three-dimensional presentation subsystem, an author- and instructor-attributed content-authoring model, a consent-gated live voice signal analysis capability, and a generalized external-connector architecture through which a separately-owned, separately-patented external fact-retrieval resource may optionally be connected without becoming a dependency of the core system.

A third correctness property is specified: determinism is asserted between two executions where and only where a context key, computed over an enumerated set of locked context dimensions, is identical. The locked dimensions are device class, entitlement set, locale, expressive register, audience level, temporal posture, traversal cost table and convergence regime. Two executions differing in a locked dimension yield differing keys and a difference between them is not a defect; two differing only in an unlocked dimension yield an identical key and a difference is a defect. Specified at §124.2.

A large language model converts input into token sequences and generates output through statistical next-token prediction; it has no native mechanism to distinguish a well-supported answer from a plausible-sounding one, because both are produced by the identical generative process. QueryBook converts input into Fact Units, and generation is constrained to vocabulary traceable to those records — an ungrounded assertion is structurally unavailable to the generation process, not merely statistically discouraged.

This distinction is implemented through a dual-architecture cognitive layer separating a Semantic Brain (work-specific, language-anchored, reversible, handling deterministic fact-anchored reasoning) from a World Brain (domain-general, language-agnostic, irreversible after a defined consolidation event, handling broader conceptual and world-state modeling) — a two-level predictive-coding hierarchy a language model's single-matrix architecture does not have.

The hierarchy extends beyond two levels. A deeper predictive hierarchy is specified, in which predictive levels are registered, a prediction error is attributed to the lowest level whose model accounts for it, an escalation gate governs propagation to a higher level, and an error attributable to no registered level is retained rather than discarded. Specified at Chapter 122.

2.1 Comparative Advantages Over Agent Frameworks, RAG Pipelines, and Chatbot Systems

(a) Traceability: every response is bound to a specific, cryptographically-verifiable set of contributing Fact Units, rather than a citation naming a source without a verifiable binding to it.

(b) Structural (not statistical) groundedness: an ungrounded output is unavailable to the generation process, as described above — a categorically different guarantee than reducing an undesired output's statistical likelihood.

(c) Governed self-improvement: the system's own improvement process requires a named, accountable human's approval at a non-bypassable gate, with a fixed set of subsystems permanently excluded from any proposal — a governance property most agent frameworks and RAG pipelines do not specify at the architectural level.

(d) Deployment flexibility: the same architecture operates in a fully cloud-based multi-tenant configuration or a fully offline, on-premises configuration reconciled through a defined synchronization layer, rather than requiring continuous access to a hosted inference endpoint.

(e) Directed, non-adopted external integration: connections to external systems — including other agent platforms — dispatch specific, declared actions only, without the connecting system acquiring the external platform's own broader authority, addressed further in Section 14 below.

Deployment flexibility at (d) comprises five topologies: multi-tenant hosted, single-tenant local, single-device, hybrid — substrate and access domains in customer infrastructure with media assets in operator infrastructure — and edge, a frequency-selected cache near a reader concentration that holds no authority and escalates where entitlement requires. Specified at §107.1.

This comparison is made at the level of architectural properties actually specified in this application. It does not include specific, unverified performance or cost claims — for example, relative compute cost or hardware requirements — consistent with this application's practice of citing only claims traceable to specified mechanisms.

A sixth comparative property is specified and is stated here because it is the one most often asked for in evaluation: measured self-assessment. The behavioural

probes specified for evaluating an external system are applicable to this system, and the trial architecture measures determinism, refusal integrity and coverage against enumerated trial spaces, reporting the uncovered residue rather than summarising it. Specified at Chapter 117.

2.2 QueryBook's Cognitive Structure

The term neuromorphic is used in this document in a specific and limited sense, and the sense is stated before the structure is described. It denotes that a mechanism's organising principle is drawn from a verified result in neuroscience or cognitive science, that the principle extracted is deterministic, and that any trained or stochastic component of the source theory is excluded rather than imported. It does not denote spiking hardware, it does not denote a neural network, and it does not denote resemblance offered in place of specification. Every correspondence claimed in this document carries the primary publication establishing the finding it invokes.

QueryBook's cognitive structure is a layered arrangement of deterministic mechanisms over a single knowledge primitive. The primitive is the Fact Unit of Section 1. The layers are five, and each is defined by what it may do to the layer beneath it rather than by the subject matter it handles.

Layer	What it holds	What it may do
Substrate	Fact Units and their lineage.	Written to only through the ingestion conversion, which applies safety evaluation, type validation and confidence calibration. No other layer writes.
Structure	The concept structure, the predicate catalogue, the event type catalogue, and the typed edges among records.	Defines what concepts exist and how they relate. Read by every layer; written by none but itself.
Convergence	Activation values over candidate records.	Resolves a candidate set to a stable configuration. Holds no records of its own and writes none.
Inference	Derived conclusions and their disposition.	May write a deductive conclusion as a record marked derived. May not write an inductive or abductive one, which is emitted in a proposed state.
Expression	A graph over stabilized records and the text realised	Composes under a vocabulary constrained to the

Layer	What it holds	What it may do
	from it.	records beneath. Writes nothing back.

The arrangement is acyclic in the direction that matters: expression cannot write to the substrate, convergence cannot write to structure, and inference writes only what follows necessarily. What flows upward is records and activations; what flows downward is nothing. This is the property that makes the groundedness guarantee structural rather than procedural, because a proposition composed at the expression layer has no route by which it could have originated anywhere but the substrate.

2.3 The Cognitive Process

A query passes through the layers in a fixed order, and the mechanisms it meets are of six distinct kinds. They are named here with the technical terms proper to each, because the terms carry the precise commitments the mechanisms make.

Associative retrieval and convergence

Candidate records semantically proximate to the query are pre-activated — priming, in the cognitive sense — within a traversal budget expressed in units of edge-class cost. The candidate set is then resolved by an attractor network: activation values are iteratively updated as a threshold function of a weighted sum until the configuration reaches a stable state, the number of stored configurations being held below the capacity at which retrieval states cease to exist. Where several interpretations compete, resolution proceeds by lateral inhibition — each candidate reduced in proportion to the summed activation of those inconsistent with it — which at a sufficient suppression coefficient is a winner-take-all network. The whole operation is deterministic; nothing is sampled.

Graph and hypergraph traversal

Relationships among records are held as typed weighted edges in a directed graph, and relationships among more than two records as hyperedges, in which a single edge relates a set rather than a pair. The distinction is not notational. A reaction with four reactants and three products, a contract with five parties, a citation drawing on six sources, are single relations over a set; decomposing them into pairwise edges loses the fact that the constituents belong together. A partially-observed hyperedge is completed by the same convergence described above, applied at hyperedge granularity and weighted by each constituent's confidence and accumulated evidence.

Geometric and spatial reasoning

Records carrying spatial anchors support reasoning over position, orientation and containment. Spatial indexing proceeds over a hexagonal tiling, adopted because it tiles a plane with uniform neighbour distance rather than because it resembles any biological structure. Geometric relations — adjacency, enclosure, occlusion, alignment — are computed rather than asserted, and a completion of a partially-observed scene is evaluated against registered physical constraints before it is scored for plausibility, a completion whose residual exceeds tolerance being absent from the output at any score. A gap is preferred to an impossible completion.

Deduction, induction and abduction

Three reasoning modes operate and are distinguished by what may follow from each. A deductive conclusion — subsumption, defeasible inheritance, transitive closure, contradiction detection — is admitted as a record marked derived, carrying a confidence equal to the product of the traversed edge confidences, because a conclusion following necessarily from admitted records adds nothing not already entailed. An inductive conclusion — accumulation, generalisation, concept formation from repetition — adjusts a confidence value within an existing record or is emitted in a proposed state. An abductive conclusion, ranking candidate explanations by posterior, sits on the same gate. Deduction may write; induction and abduction may not.

Prediction

Prediction operates at the level of semantic structure rather than linguistic surface form: what is predicted is the next relevant Fact Unit rather than the next token. Sequence prediction learns which records follow which across a session and pre-weights the predicted set; forward inference traverses causal and logical chains to conclusions no single record states; counterfactual projection represents a hypothesis as a delta over the base state rather than as a copy; and error propagation revises the concept model that generated a failed prediction at a precision-weighted gain rather than revising only the record contradicted.

Hypothesis

A conjectural derivation is created in a proposed state and transitions to validated only upon a confirmation recorded against a named human reviewer, no automated process being configured to effect that transition. Where two hypotheses stand within the separation floor while each exceeds the support minimum, the discriminating-instance mechanism selects the observation maximising the difference between its likelihood under each — not the observation either hypothesis considers most probable — and states what each posterior would

become under both the presence and the absence of that observation, which is the difference between a test and a search for confirmation.

Six kinds of mechanism, one primitive, five layers, and no path by which a proposition may reach the reader without having originated in an admitted record. That is the cognitive structure, and the sections that follow specify each of its parts.

QUERYBOOK CORE TECHNOLOGIES

3. Neuromorphic Cognitive Layer (NCL) and Retrieval Stabilization

A Neuromorphic Cognitive Layer implements retrieval stabilization through a continuous-valued associative memory network that iteratively updates an initial candidate Fact Unit activation set according to a defined convergence rule until the set reaches a stable state, resolving ambiguous queries to their best-supported interpretation as a single, deterministic convergence process. The same Neuromorphic Cognitive Layer additionally implements the Temporal Memory Module and the pre-activation sense of the Associative Recall Engine described in Section 21 below as components of the Predictive Cognitive Architecture — these are not separate systems loosely inspired by neuromorphic principles, but structural components specified within the Neuromorphic Cognitive Layer itself, unified with the broader predictive architecture in Section 21.

> **NEUROMORPHIC CORRESPONDENCE.** The convergence network is an attractor network in the sense of Hopfield (1982, PNAS 79:2554–2558), and the capacity ceiling named at §7.6.4 is the storage limit established for Hebbian-prescribed weights by Amit, Gutfreund and Sompolinsky (1985, Physical Review Letters 55:1530–1533), at which retrieval states disappear discontinuously — a first-order phase transition rather than a gradual degradation. The set of initial configurations converging upon a given equilibrium is that equilibrium's attractor basin; under the spectral contraction of §116.2 the whole configuration space forms a single basin, which is what makes the converged state unique. Weight updating follows a Hebbian rule, incrementing the weight between two records upon their co-activation, and is deterministic and auditable rather than fitted by gradient descent — a distinction stated because the term is otherwise apt to be read as implying training.

Four properties of the convergence are specified. The update decreases monotonically an energy over the configuration, so termination follows from the descent rather than from an iteration bound. The capacity at which retrieval states cease to exist is named as a first-order phase boundary, with a conservative operating ceiling below it. A network may elect a spectral-norm contraction under which the equilibrium is unique independently of the starting configuration, and an output discloses which convergence regime produced it. And the same

convergence is applied at hyperedge granularity, completing a partially-observed hyperedge weighted by each constituent's confidence and accumulated evidence. Specified at §7.6.4, §53.7, §116.1 and §116.2.

The convergence additionally admits an analog realization. The threshold update is a weighted sum followed by a sign, which a crossbar of variable-conductance elements performs physically by Ohm's and Kirchhoff's laws in a single settling; for a network of 8,192 units one digital update performs 67,108,864 multiply-accumulate operations where one settling performs a single physical operation. An analog realization is permitted only under the unique-equilibrium regime, because two arrays whose conductances differ by fabrication tolerance settle to a common equilibrium only where the equilibrium is unique. In simulation over 64 units holding six patterns, two matrices differing by five per cent conductance variation settled differently on 30.5 per cent of inputs under the capacity bound and on 0.5 per cent under a contraction. Specified at §7.7.

4. Cross-Domain Consistency

A rule engine evaluates a probabilistically-weighted, temporally-scoped rule set incrementally as new Fact Units are ingested, maintaining a persistent, queryable index of currently-active contradictions, supporting multi-hop reasoning.

A coherence classification is specified over six dimensions — semantic, temporal, spatial, editorial, media and evidential — with the governing value taken as the least of the six rather than their mean, because a mean permits five strong dimensions to conceal the one weak dimension that constitutes the finding. Classification names the dimension that produced it, and distinguishes two states from the three graded ones: incomplete, where a dimension cannot be computed for want of data, and contradictory, where two records assert incompatible content. A coherence value informs and never gates. Specified at §124.1.

5. Confidence Calibration

Each Fact Unit's confidence value is produced by a calibration process integrating extraction confidence, source reliability, a modality-specific coefficient, and temporal decay, operating upon the evidence pair of Section 1 rather than upon a scalar. Temporal decay acts upon the parameters toward the prior, preserving the reported mean while widening the credible interval, rather than reducing the reported confidence directly. A Fact Unit independently corroborated by more than one ingestion modality receives a calibrated elevation, weighted by source-class diversity so that repeated corroboration within one class contributes less than corroboration across several.

> **NEUROMORPHIC CORRESPONDENCE.** The cross-modality elevation corresponds to intersensory redundancy as established by Bahrick and Lickliter (2000,

Developmental Psychology 36:190–201): a property presented concurrently to two or more senses is detected earlier and more reliably than the same property presented to one. The scaling by temporal synchrony is drawn from the same finding — what makes two modal observations independent evidence rather than one event observed twice is that they co-occur within a window narrow enough to indicate a common cause.

6. Deterministic Provenance

Every Fact Unit's derivation is recorded in a provenance ledger implemented as a Merkle directed acyclic graph, in which each node carries a keyed cryptographic checksum sufficient to detect any operation that bypassed a required safety evaluation, enabling independent, third-party verification of any cited response.

Each node is additionally signed under a key of the deployment, so that the chain is verifiable by a party holding the corresponding verification key without access to the system that produced it. Verification establishes that a recorded operation and its recorded safety evaluation are as written; it establishes nothing about the correctness of the operation, and the distinction is stated so that an audit is not read as more than it is. Specified at §107.5.

Implementation update (prototype v9.64). The ledger described above is now exercised end to end in the prototype at two levels of assurance. Locally, every ingested artefact, sealed submission and certification event is appended to a hash-chained provenance ledger in which each seal's identifier commits to its predecessor, so that any alteration of history is detectable on recomputation. Externally, a seal may be certified on a public blockchain as described in Section 51, so that its existence and content hash can be verified by a party who does not trust the producing machine. A separate tamper-evident security record, described in Section 53, applies the same construction to security decisions. As stated above, verification establishes that a record is as written; it establishes nothing about the truth of what the record asserts.

7. Multimodal Ingestion

Content is ingested across eight defined modalities through independent, per-modality extraction processors converging on the common Fact Unit schema, with a defined fallback — best-effort resolution by structural similarity, or an explicit, stated halt — for content that does not cleanly resolve to any of the eight. The system is designed specifically to extend a static text, document, or media asset into a live, interactive question-answering capability, and is not limited to single-source ingestion: multiple documents may be ingested into a common Fact Graph, queried, compared, and cross-referenced against one another, with contradiction and temporal-inconsistency detection (Section 4 above) operating across ingested sources rather than being confined to any single one.

Two further ingestion pathways are specified. A computational substrate external to the system — a neuromorphic processor, a memristive array, a multi-electrode array — may be registered as a source and its output registered as a modality with its own independently-failing extractor; such output is admitted as a measurement of the substrate's behaviour and bears upon any other subject only through an interpretation registered in advance. Specified at §4.10. And an external machine-learning engine may be prompted and its reply ingested, which is Section 40 below.

Implementation update (prototype v9.64). The Open Claw pathway for arbitrary artefacts is now implemented and is described in Section 50. It realises the defined fallback of this section for content that does not cleanly resolve to a single modality: an artefact of any format is identified from its content, divided into addressable segments, and converted into Fact Units of the common schema, or, where no segment resolves, described rather than interpreted. Every artefact so ingested is treated as data and never as instruction.

7.1 Multi-Language and Translation Capability

Because a Fact Unit's identity is anchored to a language-independent semantic fingerprint rather than to the specific language of its source text, the same underlying Fact Graph supports query and response generation in multiple languages without separate, per-language re-ingestion. A dedicated multilingual regeneration capability generates natural-language output directly from the language-independent Narrative Graph and its underlying Fact Units for a requested target language, applying the identical structural-groundedness constraint of Section 6 regardless of output language — a translated response is subject to the same traceability requirement as a response in the content's original language, rather than being generated by translating already-generated text after the fact.

Three refinements are specified. A wording-operative span — a quotation, a statutory phrase, a claim term, a prescribed instruction — is reproduced verbatim in its source language with a separately-marked non-operative gloss, because regenerating a span that carries force alters the force. An idiom is resolved by detecting divergence between its compositional and attested readings and substituting only a registered equivalent, a surface-similarity selection being refused. And two or more language realizations may be presented simultaneously, aligned by knowledge record rather than by position, so a citation taken in either resolves to the same record. Specified at §16.7, §16.8 and §107.4.

Why this structure enables what per-language systems cannot

The capability described above follows from one decision taken at the level of the primitive, and the reason it is difficult to retrofit is worth stating plainly. In a

system that stores text, meaning is inseparable from the words carrying it, so a corpus in six languages is six corpora: a query in one reaches one, a correction must be made six times, and a divergence between two of them is discoverable only by someone who reads both. Translation between them is a text-to-text operation, and an error introduced in translation is indistinguishable afterwards from an error in the source.

Because a Fact Unit's identity is a semantic fingerprint computed over its canonicalised triple rather than over its source wording, the same assertion arriving in six languages converges upon one record carrying six provenance branches. Four consequences follow that a per-language architecture cannot produce without rebuilding around this decision.

A query in any language reaches content ingested in any other, because the query resolves to concept and predicate identifiers and the records are indexed by those identifiers rather than by words. No translation of the query occurs, and none is needed.

A correction is made once. Revising an assertion revises the record; every language rendering is generated from the record and therefore follows. Drift between language editions is not merely avoidable but detectable, because two renderings asserting different things are two records with a contradiction between them rather than two texts nobody compared.

A response in any language is generated directly from the records rather than by translating a response already generated, so the structural-groundedness constraint applies identically in every language. A translated answer is not a translation of a grounded answer — it is a grounded answer, composed under the same vocabulary constraint from the same records.

And corroboration across languages counts as corroboration across source classes rather than as repetition. The same assertion attested in an English filing and a Japanese one is evidence from two classes; in a text-based system it is two documents that happen to say the same thing, with no mechanism recognising that they do.

The cost of the decision is paid at ingestion, where an assertion must be resolved to canonical identifiers rather than stored as written, and where a wording-operative span must be identified and preserved rather than reduced. That cost is incurred once per assertion. The alternative is incurred once per language, forever.

8. License-Scoped Query Restriction

Every query is evaluated against the requesting identity's actual licensed content scope before execution, enforced at the query-execution layer itself rather than as a post-hoc filter.

The scope description constructed before traversal is bounded by four further conditions where they apply: an analysis record is excluded from a scope description constructed for a query not directed to it; a reader's progress marker excludes a record whose narrative position exceeds it where spoiler protection is elected; an attestation cutoff excludes a record attested later than the cutoff where the scope is constructed for a retrospective forecast run; and the intersection of participants' entitlements bounds a shared session. In every case the excluded record is not traversable rather than removed from a result. Specified at §96.10, §105.7, §103.3 and §105.4.

9. Live Interaction Voice Signal Analysis

Where a live, present user explicitly and revocably opts in, prosodic and acoustic features of that user's voice during an active session may be analyzed using the same extraction technique already specified for ingested audio content, applied to a live stream instead. This capability is strictly opt-in, disabled by default, and immediately halted upon revocation; it feeds an existing clarification-request mechanism as an additional, optional input and, where directly queried, is presented as an inference carrying an explicit confidence score rather than an assertion of the user's actual internal state.

A distinct voice capability is specified and is not to be conflated with this one: voice navigation accepts spoken command and query input under its own session-scoped revocable consent, routes the transcription through intent classification separating query intent from operational intent, requests clarification below the confidence threshold rather than selecting an interpretation, discards audio after transcription, derives no voiceprint or biometric identifier, and requires a second-modality confirmation for any state-changing instruction. Navigation consumes the words; the analysis of this section consumes the signal; enabling either does not enable the other. Specified at §105.6.

10. The Prime Directive

A foundational constraint, the Prime Directive, governs every subsystem described in this application uniformly: the system shall not, through any query answer, commerce signal, AI-generated application, Fact Graph content, or automated or scheduled output, cause or materially contribute to harm to a person, group of people, or the operator's organization, where that harm could reasonably have been avoided without withholding information the requester is otherwise entitled

to receive. No subsystem is exempt from this constraint, and no subsystem may be configured to bypass it. A companion rule, the Without-Information Rule, specifically prohibits withholding information a requester is entitled to receive as a means of avoiding a harm determination — the system may not resolve a difficult case by silently omitting the answer.

The Prime Directive is specified as Domain 0, a vertical domain penetrating every horizontal domain, and owns four mechanisms in addition to the constraint itself: the safety enforcement plane, the transfer matrix governing every permitted movement between domains, the path-separation invariant, and the scheduling priority ordering. Each is a member of the permanently-excluded subsystem set, and the placement is not administrative: the transfer matrix contains the entry denying the platform-governance domain a transfer to Domain 0, and were the matrix owned by that domain, the exclusion protecting the Prime Directive would be administered by the party it excludes. Specified at Chapter 106.

11. World-View Model (WVM)

A World-View Model implements the system's semantic spine, spanning ontology, causality, and time across the platform, natively and without dependency on any external or independently-operable technology. The WVM implements an ontological hierarchy, a causal graph, a temporal model, a semantic consistency engine validating non-contradiction, causal coherence, temporal alignment, and ontological validity across the knowledge graph, and versioning of the world model itself as it evolves with new ingested content. The WVM is distinct from, and complementary to, physical and embodied world modeling used for sensor and device contexts specifically — the WVM is the general semantic spine; physical-context modeling is a separate, narrower layer serving embodied applications.

Three mechanisms of the WVM are specified in detail. Attribute inheritance along IS-A chains computes confidence as the product of the traversed edges at traversal time rather than writing it back; an exception override suppresses an inherited value for a subordinate and its subordinates by specificity ordering, leaving the superordinate unmodified; and where two conflicting attributes are inherited at identical depth the system performs no resolution at all, recording the conflict and reporting both values with their originating concepts, because an arbitrary tie-break produces an answer that is reproducible without being grounded. Inverse traversal proceeds only through a declared inverse predicate, no second edge being stored. Specified at §54.6.

> **NEUROMORPHIC CORRESPONDENCE.** Two further correspondences are carried by mechanisms of this layer. Spatial indexing over a hexagonal tiling corresponds to the firing structure of entorhinal grid cells reported by Hafting et al. (2005, Nature 436:801–806), the hexagonal arrangement being adopted because it tiles a plane with uniform neighbour distance rather than because a grid cell resembles an

index. And the retention of a single object identity across a full occlusion within a session corresponds to object permanence as described by Piaget (1954): the object is not re-created as a second node upon reappearing, because it was never treated as having ceased to exist.

The physical-context layer referred to above carries a stated mechanism. A governing physical constraint is registered with a closed-form residual expression, a tolerance, SI units, a mandatory applicability predicate and its provenance; a scene completion is evaluated against every applicable constraint before plausibility scoring, and a completion whose residual exceeds tolerance is absent from the ranked output at any score. A gap is preferred to an impossible completion. An ingested observation is not refused by this mechanism, being routed instead to the contradiction index. Specified at §53.9.

11.1 Hypergraph Structure

Relations among records are held as typed weighted edges in a directed graph where a relation holds between two things, and as hyperedges where a relation holds among more than two. The distinction is not notational. A reaction with four reactants and three products, a contract binding five parties, a proof drawing on six premises, an event with several simultaneous participants — each is one relation over a set. Decomposing it into pairwise edges loses the fact that the constituents belong together, and no quantity of pairwise edges recovers it.

Two operations distinguish a hyperedge from a set of edges. Where a reasoning operation traverses a hyperedge, each constituent's contribution is weighted by its own confidence and by its degree of corroboration with the others, so a hyperedge one of whose constituents is weakly supported does not carry the strength its best-supported constituent would suggest. And a partially-observed hyperedge — a contour with observed points and gaps, a reaction with an unidentified product, a citation network with a missing source — is completed by applying the convergence dynamics of Section 3 to the constituent set as a whole rather than by inferring each absent constituent independently, because the constituents constrain one another.

A completed constituent is marked as completed, is bounded in confidence by the observed constituents and the configuration converged to, and is inadmissible as a grounding source. Where convergence fails within the iteration ceiling, no completion is produced and the gap is reported as a gap.

11.2 Concept Formation

A concept node is not authored. It is formed by extracting properties that persist across many ingested instances — observing many instances, extracting the stable properties, building a model independent of the context in which any instance

appeared, and using that model to recognise later instances. The correspondence is to invariant model formation in cortical columns, and the property being claimed is the extraction of invariance across exposures rather than any resemblance to neural tissue.

Formation is gated rather than continuous. Session-scoped records constitute a fast episodic store and the permanent Fact Graph a slow consolidated one, and a candidate concept transitions from the first to the second only upon an explicit election — never by automatic replay. The consequence is that a concept does not form because material was encountered often; it forms because someone decided the pattern was a concept. The correspondence here is to complementary learning systems as formulated by McClelland, McNaughton and O'Reilly, and the gating is what prevents the system from generalising from repetition alone.

Where a concept is grounded in more than one stream, the streams are built independently and fused only on agreement. A sensory-grounded stream and a language-grounded stream develop separately; upon co-occurrence of a spoken label and a perceived object the semantic consistency of the two is confirmed and only then are they joined by a binding edge. Fusing on co-occurrence alone would bind a label to whatever happened to be present when it was uttered.

A formed concept carries a stability weight reflecting accumulated evidence and consistency, which initialises the pseudo-count of its confidence parameters. A settled concept therefore resists revision by the arithmetic rather than by a rule expressing that intention, requiring correspondingly stronger contradicting evidence to move.

11.3 Selective Attention and Focus

Where several objects are detected concurrently, a salience weight is computed for each from its detection confidence and its relevance to the current task, normalised against the objects co-present with it. Attention is therefore relative rather than absolute: the same object is salient in one scene and not in another, according to what accompanies it.

The governing constraint is stated because it is what distinguishes this from filtering. A node is never removed from the graph, never excluded, and never rendered unreachable by a low salience weight. Salience orders what is examined first and how much traversal budget is spent upon it; it does not determine what exists. A system that discarded the unattended would be unable to answer a later question about it, and the whole point of holding the record is that the question may come.

The correspondence is to biased competition as identified by Desimone and Duncan: representations compete for processing, and a bias — here detection

confidence and task relevance — resolves the competition without a separate selector deciding among them. The same mechanism, applied to competing interpretations of a query rather than to competing objects in a scene, is the competitive resolution of Section 21.4.3.

Focus operates at a second level through traversal cost. Pre-activation is bounded by a budget expressed in units of edge-class cost, and traversal prefers directions of lower relevance gradient — regions where meaning changes slowly, rather than boundaries where it changes sharply. Attention selects what to look at; the cost table determines how far to look in each direction.

11.4 What the World-View Model does not do

Three limits are stated because each is easy to assume of a component described as a world model. It forms no concept without an election, so it does not learn continuously from exposure. It discards nothing on grounds of salience, so an unattended observation remains queryable. And where a prediction error is accounted for by no registered level of the hierarchy, the observation is recorded in the contradiction index and no revision is applied anywhere — an error no level explains is a finding about the model rather than an instruction to change it.

12. Fact Unit Example

The following illustrates the Fact Unit architecture of Section 1 applied to a concrete assertion. Given source content stating that a named entity is located within a named containing region, the ingestion pipeline resolves the subject and object references to canonical, language-neutral identifiers within the Ontology Mesh, resolves the asserted relationship to a canonical predicate identifier, and constructs a Fact Unit binding the three: a deterministically-derived FUID; the resolved semantic assertion; a confidence score reflecting extraction agreement and source reliability; a safety classification; a temporal anchor reflecting when the assertion was true and when it was ingested; and a provenance reference identifying the specific source passage. A subsequent query requesting the location of the same named entity resolves through the identical canonical identifiers, retrieves the constructed Fact Unit deterministically — not through approximate text matching — and regenerates a natural-language answer citing the specific Fact Unit and its source passage. Where source content is later revised, a new, linked Fact Unit version is created rather than the original being modified, with both versions independently queryable through the version chain.

13. Encryption and Access Control

An opt-in, passphrase-derived encryption mechanism protects an individual user's own session and interaction history such that the platform itself cannot decrypt that history without the user's own passphrase — a zero-knowledge property in the

standard security sense. Where this mechanism is active, account erasure is implemented cryptographically: destroying the associated private key record renders the encrypted history permanently unreadable without requiring individual enumeration and deletion of every underlying record. Access to platform capabilities generally is governed by a role-based portal architecture scoped to a small set of defined roles, each mapped to a defined capability set, with a separate mechanism for granting a time-scoped, context-specific capability to an identity for a bounded purpose without creating a new role or broadening that identity's general access.

Key custody is specified: an encryption key is scoped to a tenant or a deployment such that a key of one tenant decrypts no content of another, and key material is held within a hardware or equivalent custody boundary where a deployment configures one, in which case the system holds a handle rather than the key and every use is a call across that boundary. The zero-knowledge election of this section is unchanged and is not weakened by the presence of a hardware module: where it is in effect, no operator-held key exists under any configuration. Specified at §107.5.

Implementation update (prototype v9.64). Role-based access is extended to automated callers. Every agent call — whether from a QueryBook agent, an external AI agent connected through the Model Context Protocol, or an HTTP client — now passes through a single Agent Gateway that authenticates the caller by a derived secret token, applies the caller's role and the agent's registered command allow-list, and refuses replayed requests. The gateway is the first of six protection layers described in Section 52. An allow-list changes only when an operator edits the deployment's policy and restarts the system; no interface can enlarge it.

14. External System Integration — Directed Delegation

Where the system connects to an external agent platform or device — for example, a robotics or IoT platform — it does so through the generalized connector architecture of Section 8, dispatching only specific, declared actions the external platform's registered capability declaration authorizes, and acquiring no native or general control over that external platform beyond those declared actions. Where an operator additionally elects to register an external blockchain-based ownership-verification service for a specific published title's resale — an optional, per-title, publisher-initiated choice — the system acts as the requesting party to that external, pre-existing verification service; the system does not itself operate blockchain infrastructure, and no title uses this mechanism by default.

The proposition that connection confers no dependency is structural rather than asserted, in three respects verifiable against the application itself: no claim recites

an external fact-query system as an element; no transfer-matrix entry permits a domain to require an external resource in order to complete an operation, the external-integration domain's failure being contained to its own capability; and the determinism guarantee is established by the constrained-vocabulary generation and the bounded stable-state capacity rather than by any property of an external system. Specified at §107.7.

Direct command emission to physical equipment is specified for one bounded class, at Sections 32 and 33 below, and is distinguished there from the directed delegation of this section.

15. Moderation Appeals

Where content is moderated, an affected identity may submit a formal appeal referencing the original moderation action's provenance record. A narrative explanation of the original decision is generated from the same grounded generation pipeline as any other system output, routed to a human reviewer distinct from the original moderator, whose uphold, reverse, or modify decision is logged with the reviewer's identity and stated reasoning. The original decision remains in effect during the review period.

16. Data Export and Portability

A requesting identity may export the Fact Units, annotations, and interaction history they hold rights to, serialized to an open, non-proprietary format with provenance metadata preserved, without the export restricting that identity's continued access to the underlying system. The export scope is governed by the same license-scoped access-control mechanism described in Section 8.

17. Accessibility Suite

Where content is produced for delivery to a reader, a corresponding accessible equivalent — alternative text for visual content, scene descriptions for immersive presentations, captions for audio, and structurally simplified variants for cognitive accessibility — is generated from the same source Fact Units as the primary content, with production treated as incomplete until the accessible equivalent exists alongside it.

18. Offline and Synchronized Operation

The system supports a plurality of synchronization modes — including real-time, deferred, batch, and fully disconnected operation — selected according to available connectivity, with query execution, provenance verification, and access-control enforcement all continuing to function during disconnected operation using a locally-held Fact Graph. Upon reconnection, divergent local and remote states are

reconciled through a deterministic, version-vector-based merge process rather than an ad hoc or manual procedure.

19. Contextual Commerce Layer

An optional, operator-configured layer surfaces commercial content — author- or publisher-linked products, third-party affiliate or advertising-network content, or both, at the operator's choice — based on the concept identifier a query resolves to, structurally separated from the grounded answer itself. The grounded answer's Fact Unit retrieval, ranking, and confidence-score computation are completed and finalized before this layer is invoked; the layer receives only the resolved concept identifier as read-only input and has no write path back into retrieval, ranking, or generation. This structural ordering — not a configurable policy setting — is what prevents commercial content from being able to influence the grounded answer under any operator configuration. Commercial content is rendered in a distinct, source-labeled region of the response, never interleaved into the grounded answer's own sentences or citations. Every commercial-content display event is logged and operator-queryable, recording the triggering concept identifier, the specific content displayed, its source, and a timestamp.

Processing consumed by this layer is measured as its own bounded, prioritized unit of compute accounting, scheduled strictly below every safety-relevant accounting category, such that a slow or unresponsive third-party commerce source degrades only the commercial content region of a response, silently, without delaying or degrading the grounded answer. Where more than one candidate commercial item matches a query's resolved concept identifier, a selection weighting — combining match relevance, an operator-configurable source-priority factor, and, for third-party sources, a supplied bid or revenue-share value — determines which candidate is displayed, operating exclusively on candidates already gathered after the grounded answer's own completion, with the specific numeric weights configurable per deployment while the non-interference guarantee itself remains fixed.

The input to commercial selection is the resolved concept identifier together with the identifiers of concepts adjacent to it within a single edge, each supplied read-only. Adjacency is bounded three ways: one hop with no traversal from an adjacent concept; restricted to adjacency-bearing edge types, excluding causal, temporal and contradiction edges; and confined to the requester's scope description, an adjacent concept outside it being excluded without the exclusion being signalled. An adjacent concept participates in the relevance factor alone and never in eligibility, and relevance computed through adjacency is attenuated relative to direct relevance. The ordering and isolation guarantees of this section are unchanged. Specified at §113.8.

Eligibility, placement and transaction are specified. Eligibility is determined from four already-established facts about the request — registered rights, recorded territory, delivered edition, device class — and from no other input; an inferred ergonomic signal, an inferred cognitive-load indicator and an inferred affective state are unavailable to commercial selection. A commercial item occupies a registered placement class, never interleaved with propositions, bounded by a density ceiling. Every item carries a provenance record with the presenting party's verified identity, and is not a knowledge record: not citable, not traversable, contributing to no confidence value. Commerce objects carry a price per territory, an object with no registered price for a territory not being offered there; and a transaction that cannot be reconciled against an entitlement change is refused rather than recorded pending. Specified at Chapter 113.

20. Ontology Mesh

Predicates referenced by the knowledge record of Section 1, and by the Ontology Layer of the World-View Model (Section 11), are resolved against an Ontology Mesh governing concept and predicate identity across the knowledge graph.

20.1 Poly-Hierarchical Concept Structure

A concept in the Ontology Mesh may participate in multiple independent taxonomic hierarchies simultaneously — for example, a single concept may be classified as a subtype within a scientific-domain hierarchy and, separately and concurrently, as a subtype within a legal or regulatory hierarchy — rather than being confined to one fixed classification tree. This enables cross-domain reasoning a single-hierarchy classification system cannot support.

20.2 Selective Open-World and Closed-World Reasoning

The Ontology Mesh operates under an open-world assumption by default — the absence of a given fact from the knowledge graph is not itself treated as evidence that the fact is false — while specific, individually-designated sub-domains may be declared closed-world where completeness can be guaranteed and is required, such as a regulated interaction or compliance catalog. This selective mixture, rather than a single global assumption applied uniformly, is itself a deliberate architectural choice: a uniformly open-world system cannot support completeness-dependent safety checks, and a uniformly closed-world system produces false negatives across the large majority of general knowledge domains where completeness cannot be guaranteed.

20.3 Layered Concept Architecture

A fixed, version-locked upper ontology defines a small set of universal top-level categories — including, without limitation, categories for independently-existing

entities, occurrences at a time or over an interval, conditions holding over an interval, typed connections between entities, attributes an entity may possess, and temporal position or ordering — to which every domain-specific concept is ultimately mapped, regardless of originating domain. Federated domain ontologies extend this fixed upper layer with domain-specific concept types and predicates, each governed by a designated steward and registered against the upper layer before activation, rather than each domain maintaining an independent, unrelated classification scheme. Every domain ontology's concepts remain traceable to the same fixed top-level categories, enabling structured cross-domain reasoning — for example, relating a regulatory concept from one federated domain to a scientific concept from another — that would not be possible if each domain's classification existed in isolation.

20.4 Governed Predicate Catalog

Every predicate usable in a knowledge-record assertion is drawn from a governed catalog in which each predicate carries an immutable identifier assigned at registration, rather than being introduced ad hoc at ingestion time by whatever extraction process produced a given assertion. This governance discipline is what makes the poly-hierarchical and cross-domain reasoning of Sections 20.1 through 20.3 practically enforceable: without a governed, immutably-identified predicate set, the same real-world relationship could be extracted under inconsistent labels by different ingestion events, silently fragmenting what should be a single, queryable relationship into several unrelated ones.

20.5 Backward-Compatible Versioning

Changes to the ontology are versioned, with changes that would narrow or remove an existing concept or predicate's validity prohibited outside a major version change requiring full impact analysis. Every previously-created knowledge record remains valid under the ontology version in effect when it was created, and is separately eligible for re-validation against a newer version — an ontology update does not retroactively invalidate or silently reinterpret existing knowledge records.

A governed event type catalogue is specified on the same discipline: an immutable identifier assigned at registration, a poly-hierarchical placement within the mesh, a semantic definition stating both coverage and exclusions, a machine definition enumerating required, optional and forbidden fields, versioning permitting narrowing only, and deprecation rather than deletion. An environment vector is content-addressed by a condition hash and carries its own confidence, provenance and attestation time, attached through an annotation set so that no schema field is added, with correction by supersession propagating to every referencing record. Specified at Chapter 104.

21. Predictive Cognitive Architecture (PCA)

QueryBook comprises a unified Predictive Cognitive Architecture organizing prediction mechanisms that arise independently across the system's other subsystems into a single, structurally-isomorphic correspondence to an established two-part neuroscience framework: Predictive Coding, and the Thousand Brains Theory of Intelligence developed by Jeff Hawkins. This correspondence is stated as a specific architectural claim, not a general inspirational analogy: the mechanisms described below are structurally isomorphic to the referenced framework at the level of meaning representation, not merely evocative of it.

21.1 The Foundational Distinction

Predictive Coding holds that the neocortex continuously generates top-down predictions of sensory data and processes only the prediction error — the mismatch between expectation and reality — propagating that error upward through the cortical hierarchy to update the generative model. The Thousand Brains Theory extends this by proposing that the neocortex's cortical columns each function as an independent, complete modeling system, building invariant models of objects and concepts through a reference-frame mechanism, with perception achieved through democratic voting across many independently-operating column models rather than a single centralized model. A large language model predicts the next token in a language sequence — a surface-linguistic, statistical operation. QueryBook's Predictive Cognitive Architecture predicts the next relevant semantic structure — the next Fact Unit a reader or process will need — a sub-linguistic operation over structured assertions about reality, independent of the language in which those assertions were originally expressed. This distinction is the basis for the structural (not merely analogical) correspondence claimed in this section.

> **NEUROMORPHIC CORRESPONDENCE.** Every correspondence noted in this document is admitted under the methodology described immediately below, and each carries the primary publication establishing the finding it invokes. A correspondence explains why a mechanism is well-founded; it does not justify the mechanism, no claim recites a correspondence as an element, and no mechanism in this system is adopted because it resembles a biological one. Where a source theory contains a trained or stochastic component, that component is excluded rather than imported — which is why the correspondences here are to convergence dynamics, attentional competition, consolidation, redundancy and spatial coding, and not to any mechanism whose operation terminates in a sampled outcome.

A methodology governing every such correspondence is specified, and this section's correspondence is now subject to it. A correspondence is admitted only where the primary publication is verified, the correspondence is identified in a mechanism that already exists rather than invented for it, a deterministic structural principle is extracted, and any trained or stochastic component of the

source theory is excluded. A correspondence explains why a mechanism is well-founded; it does not justify the mechanism, and no claim recites a correspondence as an element. Specified at Chapter 101.

21.2 Comparative Architecture Table

Dimension	Human Neocortex	Large Language Models	QueryBook PCA
Prediction target	Sensory reality — next physical state	Next token in a language sequence	Next relevant semantic structure — next needed Fact Unit
Prediction level	Sub-linguistic — structured world model	Surface-linguistic — token statistics	Sub-linguistic — structured assertions about reality
Error handling	Prediction error propagates up the hierarchy to revise the generative model	No structured error signal at inference time	Contradiction propagates through error propagation at a precision-weighted gain
Model formation	Invariant models built across many exposures	Weights fitted once by gradient descent	Concept nodes formed from repeated observation, gated by a consolidation count
Consensus	Voting across independently-modelling cortical columns	Single forward pass	Competitive resolution among candidate records under mutual suppression, converging to a stable configuration (§21.4.3)
Language dependence	Independent of language	Bound to the training language distribution	Independent of language by semantic fingerprint

21.3 The Eight Existing Mechanisms

Two of the eight mechanisms below — the Temporal Memory Module and the pre-activation sense of the Associative Recall Engine — are components of the Neuromorphic Cognitive Layer introduced in Section 3 above, not newly-introduced systems. This section unifies them with six further mechanisms drawn from other subsystems into a single, coherent predictive architecture.

21.3.1 Temporal Memory Module (TMM)

Learns the sequence of Fact Units retrieved across a session and predicts which Fact Units are likely relevant to the next query before that query is submitted, pre-weighting predicted Fact Units in the Cognitive Context Window to reduce traversal cost when the prediction is confirmed.

21.3.2 Associative Recall Engine (ARE) — Pre-Activation Sense

Distinguished explicitly from the Adaptive Response Engine (§10.7 of this Detailed Description, response composition sense) — both are real systems sharing an acronym; this section refers only to the pre-activation mechanism. When processing a query, surveys the Cognitive Context Window and assigns elevated SessionRelevanceWeights to Fact Units semantically proximate to the current query, pre-activating them before graph traversal begins.

> **NEUROMORPHIC CORRESPONDENCE.** Pre-activation is priming in the cognitive sense: exposure to one representation raising the accessibility of related representations before they are called for. The elevation is applied to records semantically proximate to the query, and the neuromorphic property being claimed is the elevation itself, not a resemblance — an unprimed record remains reachable at ordinary cost, and priming alters the order in which a bounded traversal spends its budget rather than what is reachable within it.

Pre-activation is bounded by a traversal budget expressed in units of edge-class cost rather than by a hop count. A semantic edge costs least because proximity in meaning remains informative across distance; an editorial edge costs more because relevance decays across version boundaries; a temporal edge costs most because adjacency in time is weak evidence of relatedness and a long temporal walk accumulates noise. Within a budget of 24 units, a semantic class of cost 1 reaches 24 hops and a temporal class of cost 8 reaches 3. Costs are registered configuration recorded with the response and are never adjusted by the system from its own experience. Traversal order additionally prefers directions of lower relevance gradient, computed at traversal time and stored nowhere. Specified at §123.1 and §124.3.

21.3.3 Predictive Inference Engine (PIE) — Forward Simulation

Performs multi-hop forward inference over the Fact Graph, reasoning forward through causal and logical chains to produce inferred conclusions not explicitly stated in any single Fact Unit. Every inference step is a typed graph traversal grounded in stored Fact Units, traceable to specific source content at each intermediate step.

The disposition of an inference is governed by a stated rule. An inference whose conclusion follows necessarily from admitted records is admitted as a knowledge

record carrying a derived-inference type and a confidence equal to the product of the contributing edge confidences. An inference that is merely well-supported is withheld from admission and emitted in a proposed state that no automated process may transition to validated. Deduction may write to the knowledge store; induction and abduction may not. Specified at Chapter 108.

21.3.4 Scenario Graph System — Counterfactual Prediction

Runs forward-model projections and counterfactual simulations across the Fact Graph, representing a hypothetical as a delta over the base knowledge state rather than an independent copy.

21.3.5 ConceptStabilityWeight (CSW) — Prior Strength

Tracks how settled a World Brain concept node is, based on accumulated evidence and consistency. A high-CSW concept node resists revision from a single contradicting source, requiring multi-source contradicting evidence to revise, literally equivalent to a strong Bayesian prior requiring a large error signal to update: confidence is maintained as a Beta posterior and the stability weight maps to the pseudo-count of the two parameters, so a high-stability concept resists revision by the arithmetic rather than by a rule expressing that intention. A belief at eighteen supporting and two contradicting moves -0.0429 on a single contradiction where one at one and one moves -0.1667.

> **NEUROMORPHIC CORRESPONDENCE.** The resistance to revision described here corresponds to synaptic consolidation: a representation strengthened by repeated corroboration becomes progressively harder to overwrite. The mechanism preventing a newly-ingested contradiction from erasing an established concept is the analogue of elastic weight consolidation as formulated by Kirkpatrick et al. (2017, PNAS 114:3521–3526) for catastrophic forgetting, with the accumulated evidence count serving the role of the Fisher information that determines how strongly a parameter resists change.

21.3.6 ConfidenceScore Contradiction Detection — Prediction Error Signal

When a newly-ingested Fact Unit contradicts an existing one, the Confidence Scoring Engine flags the conflict and demotes the lower-confidence Fact Unit — a structured response to disconfirming evidence, equivalent to the prediction-error signal at the core of Predictive Coding.

21.3.7 World Brain Concept Node Formation — Invariant Learning

Extracts invariant properties across many ingested examples, building concept nodes representing generalized, context-independent models — structurally isomorphic to cortical column invariant model formation: observing many

instances, extracting stable properties, building a context-independent model, and using that model to recognize future instances.

21.3.8 Thematic Similarity Engine (TSE) — Multi-Timescale Prediction

Identifies recurring thematic patterns across a reader's entire session history and predicts which unexplored Fact Units and works are likely of interest — a lifetime-scale prediction layer, extending prediction beyond a single session's timescale.

21.4 The Four Additions

The following four mechanisms close specific structural gaps between the eight existing mechanisms above and a complete correspondence to Predictive Coding and the Thousand Brains framework.

21.4.1 Hierarchical Error Propagation (HEP)

Where a Fact Unit is contradicted during ingestion, HEP traces upward through Cross-Modal Bindings to identify which World Brain concept node attributes predicted the contradicted value, and applies a CSW reduction to those attributes — the contradiction revises the broader world model that generated the failed prediction, not merely the specific Fact Unit. The magnitude of revision is the precision ratio: the gain is the error's precision divided by the sum of the error's and the prior's precisions, precision being the inverse of the variance computed from the evidence parameters. A blurry single observation against a settled concept yields a gain of 0.0053 and is effectively ignored; a well-established contradiction against a settled concept yields 0.5000; against a weakly-held one, 0.9947. A gain below the configured floor produces no revision, and the withheld contradiction is recorded so that a pattern of individually-insignificant contradictions remains queryable.

21.4.2 Predictive Pre-Retrieval Engine (PPRE)

During the interval between a query response being delivered and the next query being submitted, PPRE uses the TMM's sequence prediction to speculatively retrieve the top-N most likely next-needed Fact Units into a pre-retrieval buffer, assigning provisional relevance weights. A confirmed prediction yields near-zero latency on the next query; a disconfirmed prediction discards the buffer and standard traversal proceeds.

21.4.3 Competitive Candidate Resolution (CCR)

This mechanism was formerly described as Semantic Consensus Voting, in which semantic domains were said to propose interpretations in parallel and a consensus interpretation to be selected among them. That description is withdrawn and replaced here, and the original wording is preserved in the note following this

subsection so that the change is visible rather than silent. The replacement is not a reduction in scope. It states the mechanism that actually performs competitive resolution in this system, which is a closer structural match to the framework of Section 21.1 than the description it replaces.

Where a query admits several mutually-inconsistent interpretations, the candidate knowledge records supporting each interpretation are activated together rather than evaluated in sequence. Each candidate's activation is then reduced in proportion to the summed activation of the candidates inconsistent with it, and the set is iterated to a stable state under the threshold update already specified. A candidate falling below the retention floor is suppressed; the surviving configuration is the interpretation the records best support.

By way of example, three mutually-inconsistent candidates activate at 0.82, 0.61 and 0.44 under a suppression coefficient of 0.35. The first is reduced by 0.35 of the sum of the others to 0.4525; the second by 0.35 of the sum of the first and third to 0.1690; the third falls further still. The first survives the floor and the others do not, and the interpretation it supports is the one composed. Where two candidates survive at comparable activation, the set does not resolve and the clarification mechanism is invoked rather than the higher of the two being taken.

> **NEUROMORPHIC CORRESPONDENCE.** The suppression described above is lateral inhibition in the neurophysiological sense: a unit’s activation reduced in proportion to the activation of its competitors, so that the population settles upon one interpretation rather than a blend of several. Where the suppression coefficient is high enough that exactly one candidate survives the retention floor, the mechanism is a winner-take-all network. The correspondence to attentional selection is that identified by Desimone and Duncan (1995, Annual Review of Neuroscience 18:193–222) as biased competition: representations compete, and a bias — here the accumulated evidence behind each candidate — resolves the competition without a separate selector deciding among them.

Three properties follow, and each is why this is the correct mechanism to claim rather than the one it replaces.

The competition is among candidates, not among domains. In the framework of Section 21.1 what votes is a population of independent models each forming a complete hypothesis about the same object. In this system that population is the candidate record set, each candidate supporting a complete interpretation of the query. Domains are authority boundaries in which exactly one domain owns each decision — retrieval is owned by one domain, meaning by another, entitlement by a third — and a mechanism in which two domains proposed competing answers to the same question would contradict the partition rather than implement the correspondence.

The resolution is parallel and the mechanism already exists. Candidates are activated together and suppressed simultaneously rather than traversed in sequence, which is the property the withdrawn description was reaching for. It is performed by the convergence of Section 3 together with the mutual suppression specified for object salience, viewed from the predictive-architecture side; it is not a separate system, and this subsection introduces no mechanism not already specified elsewhere in this application.

The resolution is deterministic. Suppression is a fixed arithmetic operation over the activations, the coefficient and the retention floor are registered configuration recorded with the response, and the iteration terminates by the energy descent of Section 3. The same candidate set under the same configuration resolves identically on every execution, and no consensus is taken by sampling, weighting by chance, or tie-breaking arbitrarily — an unresolved tie produces a clarification request, which is a stated outcome rather than a selection.

The description replaced here read that semantic domains ran their traversals in parallel and a consensus interpretation was assembled weighted by each domain’s confidence in its own contribution. That description was not carried forward for a reason of architecture rather than of wording. A domain in this system is an authority boundary within which exactly one domain owns each decision, and a mechanism in which several domains proposed competing resolutions of a common query would require two domains to own the same decision — which the partition of Section 22 exists to prevent. There is accordingly no voting among domains in this system, and none is claimed.

What the replaced description was reaching for is nonetheless real, and is stated above: resolution among candidates is parallel rather than serial, and it is settled by competition rather than by a selector choosing between alternatives. Describing that competition at the level where it occurs — among candidate records — rather than at the level of domains yields a closer correspondence to the framework of Section 21.1, since what votes in that framework is a population of independent models each forming a complete hypothesis about the same object, and in this system that population is the candidate set.

21.4.4 Active Inference Query Generation (AIQG)

Where the PIE's forward inference chain reaches a node with confidence below a threshold, rather than stopping or returning a low-confidence result, AIQG generates a structured internal query targeting the specific Fact Unit that would resolve the uncertainty, presenting it to the reader as a targeted, purposeful question.

A distinct and complementary mechanism is specified for a case this one does not address. AIQG resolves uncertainty about a single proposition. Where two

hypotheses are supported about equally — posteriors within a separation floor while each exceeds a support minimum — gathering more evidence of the kind already held raises both and separates neither. The discriminating instance mechanism computes, for each obtainable observation, the magnitude of the difference between its likelihood under each hypothesis, and selects the maximum irrespective of how probable either hypothesis considers it: an observation both predict at 0.90 and 0.85 discriminates 0.05 and is rejected, where one predicted at 0.88 and 0.20 discriminates 0.68 and is selected. The request states what each hypothesis would become under both presence and absence, which is the difference between a test and a search for confirmation. Specified at §97.8.

21.5 The Three Amplifications

(a) Cross-Session TMM Extension — extends TMM's sequence learning across all of a given reader's prior sessions against a given work, yielding lifetime-scale sequence prediction rather than single-session prediction.

(b) Fully Bayesian ConfidenceScore — confidence updates as a proper Bayesian posterior, each new piece of evidence updating the probability distribution over possible truth values, rather than a simpler additive rule.

(c) Pre-Computed PIE Inference Chains — where the PIE runs speculatively at ingestion time, computing and storing forward inference chains a newly-committed Fact Unit enables, so that a later query triggering those inferences finds them already assembled.

21.6 Integration Requirements

This section requires Chapter 7 (TMM, Associative Recall Engine), Chapter 10 (PIE), Chapter 11 (Scenario Graph System), and the World Brain / Semantic Brain dual-architecture layer. It modifies none of their specifications — it is a formal, unifying organization of mechanisms already independently specified, plus four genuinely new additions and three genuinely new amplifications, each described above.

22. Domain and Layer Architecture

The system is partitioned into fifteen domains: one vertical and fourteen horizontal. A domain is an authority boundary rather than a deployment unit — two domains may execute in one process and one may span several — and what the partition fixes is which mechanism owns which decision, which transfers between domains are permitted, and how compute is accounted. The partition is invariant across every deployment topology.

Domain	Mandate
D0 · Prime Directive	Vertical governance. Owns the safety enforcement plane, the transfer matrix, the path-separation invariant, the scheduling priority ordering and the permanently-excluded set.
D1 · Ingestion and Sensory Intake	Conversion of external content in any modality into candidate records. The sole location of any trained component.
D2 · Knowledge Substrate	Custody of records and lineage. The only domain with write authority over the store.
D3 · Ontology and Conceptual Structure	What concepts exist and how they relate. Governs meaning rather than content.
D4 · Retrieval and Response Composition	Answering a bounded query. The latency-critical path.
D5 · Inference, Prediction and Calibration	What the records imply, and how well it was derived.
D6 · World Model and Perception	How the world behaves, held distinctly from what sources asserted.
D7 · Grounded Expression	Rendering under the vocabulary constraint. Holds no retrieval authority.
D8 · Access, Rights and Retention	What a requester may reach, for how long, and who holds the keys.
D9 · Interaction Surfaces	Presentation. No authority over content, retrieval or rights.
D10 · Authoring, Editions and Publication	Admission and shaping of authored work; emission in distributable form.
D11 · Platform Governance and Evolution	Change under human authority. Structurally forbidden from changing D0.
D12 · External Integration and Analysis	Reach to systems outside the boundary. Output quarantined.
D13 · Commerce and Entitlement Tokens	Commercial presentation and entitlement. Lowest scheduling priority.
D14 · Learning and Assessment	Assessment from grounding records, mastery measurement, credential issuance.

The functional domains group into four layers — Knowledge (D1, D2, D3, D6), Reasoning (D4, D5, D7), Experience (D8, D9, D10, D14) and Platform (D11, D12, D13) — with D0 vertical. A layer carries no authority; moving a domain between

layers alters nothing a transfer permission governs, and this is stated because a layer model treated as a permission model is the defect that renders such a model unusable.

22.1 Path Separation

Answering a query traverses D8, D4, optionally D5, then D7. Ingesting content traverses D1, D2, D3, D6, with D5 pre-computing. The intersection consists of D0 and D2 alone, so an ingestion operation of any size contends with an interactive query at no other domain. Contention at D0 is resolved by category priority; at D2 by storage tiering. Two consequences constrain implementation: D4 performs no ingestion work, so a retrieval never blocks on an extractor, and D1 holds no retrieval authority, so an extraction never blocks on convergence.

22.2 The Transfer Matrix

Every ordered pair carries a permission — permitted, denied, or permitted subject to a stated conversion — evaluated at each crossing, with an unlisted pair treated as denied. The denials carry the guarantees.

From → To	Permission	Guarantee
D12 → D4	Denied	An external system's output and any analysis of it can never enter retrieval, excepting a reply converted under Section 40.
D13 → D4; D13 → D7	Denied	Commerce cannot influence retrieval or generation under any configuration.
D9 → D2	Denied	No presentation surface reads or writes the store directly; it goes through D8 and D4.
D11 → D0	Denied	The mechanism that changes the system cannot change the Prime Directive.
D7 → D2	Denied	Generation cannot write. Output is composed from records and never becomes one.
D14 → D13	Denied	A learner's assessment record cannot reach commercial selection.
D1 → D2	Conversion	Nothing enters the store

From → To	Permission	Guarantee
		without safety evaluation, type validation and calibration.
D5 → D2	Conversion	A derived inference persists only marked as derived.
D0 → any	Permitted	Governance reaches everywhere. The definition of a vertical domain.

23. Reasoning Mode Formalization

Three modes operate, differing in what may follow from their output. Deduction — subsumption, inheritance, transitive closure, contradiction detection — produces conclusions admissible as knowledge records, because a conclusion following necessarily from admitted records adds nothing not already entailed. Induction — evidence accumulation, generalization, concept formation from repetition, discriminating-instance selection — adjusts a confidence value within an existing record or emits a proposal in a proposed state. Abduction ranks candidate explanations and sits on the same gate as induction.

Inheritance is defeasible, which classical syllogistic cannot express, and equal-depth conflicts are not resolved but recorded, because an arbitrary tie-break produces an answer reproducible without being grounded. Negative instances are retained deliberately: a contradiction too weak to revise is recorded so that a pattern of individually-insignificant contradictions remains queryable.

24. The Interpretive Layer

Five modulations — context, perspective, ergonomic, cultural and social — apply to an already-admissible result. None widens or narrows a scope description, admits or excludes a record, alters a confidence value or extends the permissible vocabulary; a modulation that would is refused. Two requesters whose scope descriptions admit identical records receive answers grounded in identical records however differently their expression is modulated.

Perspective surfaces different relationships over the same records — a student sees learning pathways, an author revision pathways, a publisher rights pathways. Ergonomics modulates density and layout, with a stated preference always overriding an inferred signal. Cultural modulation proceeds from a declared profile and never from a name, network address, query language, device locale or behavioural signal. Social modulation operates within the intersection of participants' entitlements. Composition is deterministic and Prime Directive review

precedes and follows it. **The cultural norms, social values, and ergonomic preferences these modulations apply are stored as Contextual Value Models — registered, versioned, declared-never-inferred configuration records that the interpretive layer alone reads — and are held apart from Fact Units and from World-View Model concepts: a fact about a culture is a Fact Unit, a model of how customs relate is a World-View Model concept, and a Contextual Value Model is neither, being only the declared preference under which a grounded answer is voiced, so that a reader's or an operator's values never become, by storage alone, an assertion the system treats as true; context proper is carried separately by the context lock's dimensions. Specified at §109.9.**

25. Expressive Register and Surface Realization

A response carries exactly one expressive register from a governed set — formal, plain, academic, instructional, narrative, supportive, authoritative — resolved before generation. A register governs voicing and a confidence value governs what is asserted: the authoritative register cannot remove a hedge the evidence requires, and the supportive register cannot soften an assertion the records state plainly. The supportive register is selected only on a stated preference, a configured default, or an existing consented signal, never by inference from word choice, punctuation, length, error rate or timing.

Five realization operations produce the surface text. Aggregation combines propositions sharing a subject into one coordinated sentence, refused where their confidence values differ beyond tolerance so that coordination cannot promote a weakly-supported proposition. Referring expressions take full form at first mention and reduced form thereafter, with full form restored where a same-type entity intervenes. Connectives are emitted from the graph's own edge types and never invented. Ordering places established material before new. Lexical variation is admitted only where the semantic fingerprint is preserved. Selection among admissible realizations is deterministic, never sampled.

Media assets may be carried within a response, included only where an admitted record references them and delivered with that record's citation, so no asset conveys what the text could not assert.

26. Outcome Feedback and Operational Calibration

A delivered response is recorded with its query, scope description, cited records and their confidence at delivery. A report bearing on it is admitted as a knowledge record attributed to the reporting identity: feedback is evidence, not correction. A report against a cited record increments that record's contradicting parameter by the reporting source's weight; it does not set the confidence value, supersede the

record or remove it. A record at $\alpha = 18$, $\beta = 2$ reading 0.9000 moves to 0.8633 on a report from a source of reliability 0.85. The reporting identity is a source class, so many reports from one identity do not weigh as reports from many.

Calibration is measured as squared error against realised outcomes and as a gap per confidence band, reported per domain rather than in aggregate. A finding is emitted as a proposal requiring a named human approval; nothing applies automatically, and the parameters governing the measurement are themselves excluded from modification. No feedback extends the permissible vocabulary or alters a scope description.

27. Temporal Prediction Calibration

Three timestamps are distinguished: when an assertion was true, when it was ingested, and when its source was created. The third governs admissibility to a retrospective run: a record is admissible to a pre-event snapshot only where its attestation precedes the event by the configured horizon, the temporal anchor not being consulted, and an unknown attestation excluding the record rather than being estimated. Enforcement is at scope construction rather than by post-filtering, and a contested date resolves to the latest defensible, which excludes.

Retrospective scoring is a squared-error measure with accuracy reported by horizon and calibration by confidence band. The subject class is restricted to institutions, systems and organisational processes; a subject resolving to an individual natural person is refused at construction. Aggregate statistics are perpetual; individual records follow the ordinary erasure policy.

28. Behavioural Probing and Analysis Quarantine

Three probes evaluate an external system: response consistency by output fingerprint, self-consistency under reinjection classified as converged, periodic, drifting or unresolved, and perturbation sensitivity using inputs whose fingerprints are provably identical. Every probe result requires this system's own reference value or is recorded incomplete.

Analysis records and probe results are excluded from a scope description at construction rather than filtered afterward, so an assertion this system found unsupported cannot ground a later answer. The probes are applicable to this system itself, and applying them inward measures the properties this specification claims rather than a proxy for them.

29. Systematic Trial Architecture

Trials are metered as bounded compute-accounting units in a trial category, scheduled below safety evaluation, below requester-serving and below ingestion,

with an over-budget trial deferred rather than served at reduced priority. Trials are free of physical consequence but not of cost; the cost shifts from equipment and risk to compute and scheduling.

A trial executes against an isolated corpus, constructs no production record, adjusts no production confidence value, emits no command to physical equipment outside simulation and delivers no output to a requester. Nothing a trial finds is applied: a finding becomes a proposal requiring a named human approval, and a campaign of ten million trials produces the same class of output as a campaign of one. Regression blocks rather than reports — a case passing before a change and failing after it withholds the change. Coverage is measured against an enumerated trial space with the uncovered cases listed rather than summarised. Determinism is measured over an unchanged corpus state, any value below unity being a defect rather than a tolerance. A simulated pass is not clearance to operate equipment, and where a simulation and a physical trial disagree the physical trial governs.

30. Programmatic Interfaces, Plugins and Versioning

Nine interfaces are exposed — content, query, rights, layout, assessment, edition, semantic, media and immersion — each resolving a scope description before traversal, admitting no parameter that widens one, and carrying its own safety evaluation gate as an operation layer. Calls are metered per calling identity, an over-budget call being refused rather than throttled, so a caller cannot displace a safety-evaluation unit by volume.

Each interface carries a version. A request omitting a version is refused rather than defaulted, so a caller's behaviour cannot change because a default moved. A version is deprecated across a support interval and then retired with a successor named, never deleted, and a record returned under one version remains valid under a later one.

A plugin registers against a declared capability set and reaches knowledge only through the interfaces, within the entitlement of the identity on whose behalf it runs. It holds no direct access to the record store or the provenance chain, is refused at registration where it declares any safety operation, cannot participate in the improvement pipeline, and its output is labelled with its identity, quarantined, and never citable as grounding. Activation and suspension require a named human operator.

31. Interaction Surfaces

Collaborative reading presents a shared view comprising the intersection of participants' scope descriptions, never the union; a record traversable by one participant and not another is absent, and its absence is not reported. Augmented presentation anchors records to physical objects, excluding person and animal

classifications entirely at any confidence and capturing no gaze, dwell, head-pose or attention signal. Spoiler protection bounds the scope description by a progress marker so a later-disclosed record is not traversable, rather than masking a rendered answer — a masked answer still discloses that content exists and how much.

Learner progress records state what was done and how it scored, with instructor visibility bounded by a time-scoped grant, and generate no aptitude inference, forecast or comparative ranking. Certification requires both a measured outcome and an established identity, carries its own provenance, is verifiable without disclosing the holder's responses, and attests to the measured outcome and nothing further. Learning-system integration transmits an outcome alone, with no record, response text or history.

An interactive session over a work operates in a registered mode — question and answer, discussion, tutorial, and quiz among them — and the system auto-generates the session’s materials from the work’s own records: sequenced instruction with auto-generated comprehension checks drawn from the same Fact Graph as ordinary reading; quiz and test items bounded to their session mode; flashcards generated from a topic’s records, prompt and answer sides each citation-backed; and glossaries, study guides, and spaced-repetition schedules assembled under the study-and-learning tools of the specification. Probable questions about a work are generated at ingestion together with their fully-cited answers and invalidated upon revision of any contributing record. Every generated item is composed under the vocabulary constraint and carries its citations, and an assessment item generates no aptitude inference beyond the measured outcome already specified. Specified at §52.13 and Chapter 99.

32. Deployment Topology and Module Replacement

Five topologies operate the same logical behaviour: hosted, local, single-device, hybrid and edge. A topology determines where a domain executes and never which domain owns a decision. An edge node holds no authority, answering from cache only where the requester's scope admits every cached record consulted and escalating otherwise, and performing no ingestion.

A domain's implementation may be replaced where the replacement satisfies every interface at each supported version, every transfer-matrix entry in which that domain appears, and a named human approval. D0 is not replaceable, being a member of the permanently-excluded set.

33. Directed Capture Equipment Control

This section states a capability the original Section 14 did not contemplate, and the boundary it observes. Command emission is permitted to registered equipment of a capture and inspection class — pan-tilt-zoom heads, lighting, bounded-volume capture drones, inspection arms, positioning stages, mobile capture bases — each registered with its kinematic envelope, velocity and force limits and operating volume. Emission to any other class is refused, and the prohibition on any control interface to a vehicle system is unchanged: a mechanism whose failure can injure a person at speed is not brought within scope because a mechanism whose failure spoils a photograph has been.

Command authority is granted per session rather than per command, bounded by expiry, revocability before the next command, scope to named units and volumes, and recording. Every command is evaluated before emission against operating volume, kinematic envelope, velocity and force, clearance, human proximity, inter-unit swept volume and recording-restricted zones; a violating command is withheld unconditionally, not emitted at reduced priority nor modified into a permissible one. Person detection is safety-only: used to compute clearance and for nothing else, retained beyond no evaluation, entering no result set and available to no other mechanism. A halt is evaluated from equipment state and constraints without traversing retrieval, stabilization, generation or modulation, because a stop that waits for an answer is not a stop.

35. Attractor Stabilization Beyond Retrieval

A partially-observed sensor frame may be completed by converging its observed regions to the nearest stored configuration. Three constraints bound this, because reconstruction produces content nobody observed: a reconstructed region is marked and never presented as observed; its confidence is bounded by the lesser of the observed and stored confidences; and it is inadmissible as grounding. Where convergence fails, no reconstruction is produced and the gap is reported — a frame with a hole is truthful, a frame whose hole has been filled with a guess presented as observation is not.

Alignment across editions converges only below a divergence threshold, preserving at or above it and preserving anything unclassifiable, because erasing a real editorial difference costs a reader the fact while preserving a spurious one costs a footnote. Control-signal oscillation is damped, with the damped command evaluated against constraints identically to an undamped one, and persistent oscillation raising a fault and halting rather than being damped indefinitely, because a damped symptom is a concealed fault.

36. Differentiated Propagation and Tri-Temporal Scope

Traversal cost by edge class is described at Section 21.3.1 above. Three temporal postures are distinguished and a response states which produced it: attested past, resolving under the attestation cutoff; live present, reading the live layer over the committed layer; and projected future, carrying a credible interval. Past and present results are ordinary knowledge records. A projected result is never admitted as a record and never grounds a subsequent answer. A response drawing on more than one posture carries the posture with each proposition, so a difference between two answers reads as a difference of posture rather than an inconsistency.

37. Coherence and the Context Lock

Coherence classification is described at Section 4 above. The context lock enumerates the dimensions whose values must agree for the determinism guarantee to be asserted between two executions, computes a key over them, and returns it with the output. Identical keys mean the outputs must be identical and a difference is a defect; differing keys mean nothing is claimed. The enumeration is a member of the permanently-excluded set, because a mechanism able to unlock a dimension could restore an apparent determinism by narrowing what determinism means.

38. Observation Diversity

The evidence parameters record how much evidence stands behind an assertion and not how varied it is. Each contribution is attributed to a registered source class — a distinct work, edition, language realization, modality, sensor, connector or supplemental submission — and diversity is the effective number of contributing classes, computed as the reciprocal of the concentration of contributed weight. Twenty units of weight in one class yield 1.0000; the same twenty spread across five classes yield 3.8462. Bacon's method requires varied instances rather than merely numerous ones, and a systematic error in a single source produces twenty mutually-agreeing confirmations and no evidence at all about the world.

Corroboration from a class already dominating a parameter is admitted at reduced weight. A generalization requires a diversity floor as well as a count floor. Precision-weighted revision is attenuated where a contradiction's diversity approaches unity. And a response resting on a record whose diversity is near unity states that its support is concentrated in one source class.

39. Computational Substrate Output as an Ingested Modality

A computational substrate external to the system — a neuromorphic processor, a memristive array, a multi-electrode array — is registered as a source with a

declared capability, and its output registered as a modality with an independently-failing extractor. Output is reduced to concept-level records before admission, retains no raw episode, passes the safety gate and is calibrated against the substrate's registered reliability, the substrate constituting a source class.

A substrate output is evidence of what the substrate produced and not of what it was asked about. It bears upon any other subject only through an interpretation registered in advance with its own provenance and reliability, and absent such an interpretation is admitted as a measurement of the substrate and nothing further. Derived confidence is bounded by the lesser of the substrate's and the interpretation's reliability, a reading being no better than the weaker of the instrument and the reading of it. Where a substrate comprises living tissue, the operating institution's governance and any review its jurisdiction requires apply; this specification authorises no substrate's operation.

40. The AI-Engine Ingestion Module

This section states a capability the original Section 14 expressly excluded, and the constraints under which it is admitted.

A prompt is composed by the regeneration mechanism from knowledge records under the vocabulary constraint, according to a profile registered for the target engine, and carries semantic markers naming the concepts to address, extraction instructions specifying the return form, and alignment constraints stating the target record schema. Composition is deterministic, so an elicitation may be repeated exactly and a difference in the reply attributed to the engine rather than to the asking. The prompt is dispatched through a connector under a declared capability, rate-limited and batched under compute accounting.

The reply is normalized, segmented into individual assertions, and converted into knowledge records of the ordinary schema, each receiving a temporal anchor from the elicitation, a semantic fingerprint, and a provenance reference identifying the prompt, engine, version and position within the reply. Converted records enter the ordinary store and the ordinary concept structure, are traversable, retrievable, citable, and subject to every mechanism governing any other record.

Three constraints attach, each preventing a specific failure. First, the derivation is a field of the record identifying the engine, unmodifiable and unremovable, and a citation resolving to such a record discloses it — preventing a reader from following a citation and discovering that the authority is that a model produced it when asked. Second, each engine is one source class, so three assertions from one engine yield a diversity of 1.0000 against 3.0000 from three engines, and a corroboration from the same engine is refused — preventing confidence manufactured by repetition. Third, confidence is bounded by the engine's registered reliability and, where an extraction instruction applies, by the lesser of

the two — preventing a generated assertion from outranking a reported one by fluency.

A citation contained in a model reply is verified against the store and recorded as verified or unverified, the assertion being admitted carrying that record rather than withheld, so a fabricated citation becomes a recorded property rather than an undetected defect. The audit chain records prompt, engine, version, elicitation time, position, derivation marking, the confidence bound and its basis, and the verification outcome, and is queryable.

The cost is stated rather than minimised. Before this module, every record originated from a source that asserted something about the world. With it, an answer may trace to an assertion a model generated in response to a prompt this system composed — recorded and auditable at every step, and disclosed at the point of citation, but produced rather than reported. The module is elected per deployment, and a response indicates whether any record it relied upon was model-derived.

Implementation update (prototype v9.64). Content admitted from outside the system is now screened for instruction-like text before conversion. A segment that reads as an instruction to an AI system — an attempt to override prior instructions, to reveal secrets, to assume a privileged role, or text concealed by invisible or tag characters — is admitted only as data, flagged as untrusted, and assigned a confidence below the answer threshold, so that it can neither be followed by an agent nor cited as an answer. The screening is that of Section 52.

41. Exemplary Embodiments

The following illustrate the operation of claimed methods in representative use, each tied to the specific mechanism it demonstrates rather than introducing new subject matter. Embodiments 1 through 6 are as originally drafted. Embodiments 7 through 14 illustrate mechanisms specified since.

Embodiment 1 — Single-Source Question and Answer

A reader poses a natural-language question about a specific ingested work. The system parses the query, retrieves candidate Fact Units via the mechanism of Section 3, stabilizes the candidate set through the Attractor Network's convergence process, and composes a response through the grounded generation pipeline of Section 6. The delivered answer cites the specific Fact Unit or Units it was assembled from, each independently verifiable through the provenance mechanism of Section 6.

Embodiment 2 — Multi-Source Survey With Contradiction Detection

A researcher ingests a plurality of documents addressing a common subject and poses a query spanning all of them. Retrieval proceeds across the unified Fact Graph populated by the ingestion method of Section 7, and the cross-domain consistency mechanism of Section 4 surfaces a genuine factual conflict between two of the ingested sources as a queryable contradiction-index entry, rather than silently selecting one or averaging them.

Embodiment 3 — Directed Delegation to an External System

An operator registers an external device platform through the connector architecture of Section 14, declaring a specific, named action the platform is authorized to perform. A subsequent natural-language instruction compiled through the system's bounded program layer dispatches that specific declared action to the registered platform. The system does not thereby acquire the platform's own broader capabilities — only the declared action is available to be dispatched, consistent with the directed-delegation limitation.

Embodiment 4 — Edition Comparison

A reader holding access to two editions of the same underlying work requests a comparison. The system retrieves the Fact Units associated with each edition's version chain and surfaces the specific additions, removals, and revised assertions between them, each difference traceable to a specific Fact Unit version rather than a generic text-diff of the surface content.

Embodiment 5 — Role-Scoped Portal Access

A reader, an author, and an instructor each access the same underlying published work through the role-based portal architecture referenced in Section 13. The reader's session is scoped to consumption and personal annotation; the author's session additionally exposes drafting and rights-management capability over their own published work; the instructor's session — a time-scoped capability grant rather than a distinct account type — additionally exposes class-scoped assessment and progress capability.

Embodiment 6 — Standards-Conformant Reader Deployment

The system of the preceding claims is deployed as a reading application conformant to the EPUB Reading Systems 3.3 specification, rendering content authored to the EPUB 3.3 core specification. QueryBook-specific interaction points are carried as custom data attributes on otherwise-valid EPUB 3.3 markup, ignorable by any non-conformant-aware reading system. A reader's selection of an attribute-anchored passage triggers retrieval through the mechanism of Claim 6, with the resulting output displayed through the grounded generation mechanism

of Claim 8 — the same underlying architecture operating identically whether accessed through a conformant reading system or through the system's own interface.

Embodiment 7 — A Settled Assertion Meeting a Single Contradiction

An assertion has accumulated eighteen units of supporting evidence and two of contradicting evidence, reporting a confidence of 0.9000 with a credible interval of approximately ±0.1283. A single contradicting source of reliability 1.0 is ingested. Under the additive rule this system previously used, the confidence would fall by 0.05 irrespective of what stood behind it. Under the mechanism of Section 21.5, it falls by 0.0429 — and a provisional assertion at one unit each side, meeting the identical contradiction, falls by 0.1667. The settled belief resists and the provisional one moves, by the arithmetic rather than by a rule expressing that intention.

Embodiment 8 — A Contradiction Too Weak to Revise, and Retained

A single blurry observation contradicts an attribute of a well-established concept node. The precision of the error is low and the precision of the prior is high, so the revision gain computes to 0.0053 and falls below the configured floor: no revision occurs. The contradiction is nonetheless recorded, so that a pattern of individually-insignificant contradictions against the same attribute remains queryable and a reviewer examining the concept later finds them rather than finding nothing.

Embodiment 9 — Twenty Confirmations From One Source, and From Five

Two assertions each carry twenty units of supporting weight and report an identical confidence. The first accumulated its weight from repeated readings of a single edition; the second from two editions, a translation, an image and a video. The diversity measure of Section 39 computes 1.0000 for the first and 3.8462 for the second. A response resting on the first states that its support is concentrated within one source class. Without the measure the two would be indistinguishable, and a systematic error in the single edition would present as twenty independent confirmations.

Embodiment 10 — Two Hypotheses the Evidence Cannot Separate

Two candidate explanations of a common subject stand at posteriors of 0.7500 and 0.6875, each resting on sixteen observations. The separation of 0.0625 falls below the configured floor while both exceed the support minimum, so the pair is marked undiscriminated. Three obtainable observations are evaluated: one predicted at 0.90 and 0.85 discriminates 0.05; one at 0.55 and 0.50 discriminates 0.05; one at 0.88 and 0.20 discriminates 0.68. The third is selected — not because either hypothesis considers it likely, but because they disagree about it — and the

request states what each posterior would become under both its presence and its absence.

Embodiment 11 — A Query in a Shared Session

Two readers occupy a shared session over a common work. The first's entitlement admits 340 knowledge records; the second's admits 275 of those. The shared view comprises the intersection, 275, and a record traversable by the first and not the second is absent from it — and its absence is not reported to either, because reporting that something is withheld discloses precisely what the entitlement exists to protect. A query submitted in the session resolves against the intersection and is delivered identically to both, so neither receives a citation the other cannot open.

Embodiment 12 — A Command Withheld at an Equipment Boundary

A registered inspection arm operates under a session authorization naming its unit, its volume and its permitted task classes. A compiled plan would bring the arm to within 0.42 metres of a detected person where the registered exclusion distance is 0.60. The command is withheld before emission — not emitted at reduced priority, not emitted with a warning, and not modified into a permissible command by the system's own selection. The detection of the person is used to compute the clearance and for nothing else: it is not retained beyond the evaluation, enters no result set, and is available to no other mechanism.

Embodiment 13 — An Answer Elicited From an External Engine

A deployment has elected the module of Section 40. A prompt is composed from knowledge records under the vocabulary constraint according to the profile registered for a named engine, dispatched under a declared capability, and the reply segmented into individual assertions. Each becomes a knowledge record carrying a derivation marking naming the engine, a confidence bounded by the engine's registered reliability of 0.88 and the extraction instruction's 0.62 — hence 0.62 — and a provenance reference identifying the prompt and the position of the assertion within the reply. A citation contained in one assertion resolves to no record held and is recorded as unverified, the assertion being admitted carrying that record. A later answer citing the record discloses that its authority is a named engine responding to a recorded prompt. Three further elicitations of the same assertion from the same engine raise the accumulated evidence and leave the diversity measure at 1.0000.

Embodiment 14 — Two Answers That Differ Legitimately

The same query is submitted twice over an unchanged corpus. The first is delivered under the plain register to a requester on a desktop class device; the

second under the academic register to the same requester. The two responses differ. Each carries a context key computed over the enumerated locked dimensions, and the keys differ because the register is a locked dimension — so the difference is a difference of context and not a defect. Had the two differed while carrying an identical key, the difference would be a defect, and the trial architecture of Section 29 measures exactly this.

42. Comparison With Human Cognition

Section 21.2 compares this architecture with a language model and with the neocortex on the dimensions where the correspondence of Section 21 is claimed. This section compares it with human cognition more broadly, and does so in both directions. It is included because a comparison stating only where a system is superior is not a comparison, and because the properties on which human cognition is plainly superior are the properties a reader will supply for themselves if the document does not.

Dimension	This system	Human cognition
Substrate	Software over conventional or, under §7.7, analog hardware; explicit boundaries between mechanisms.	Neural tissue; modular in function without explicit boundaries, and not decomposable into named components without loss.
Computation	Deterministic traversal, threshold convergence and constrained composition. No component is fitted by gradient descent, and no operation terminates in a sampled outcome.	Event-driven spiking, massively parallel and asynchronous, with temporal and rate coding over continuous dynamics.
Learning	Structured and bounded. A record is admitted, a confidence value moves, a concept consolidates on a count. Every change is recorded and reversible, and no change to the system itself takes effect without a named human approval.	Continuous lifelong plasticity driven by experience, reward and social feedback. No global controller and no rollback.
Memory	Explicit. A record is addressable, its origin verifiable, its confidence stated, its erasure exact.	Associative and reconstructive. Recall alters what is recalled, and the alteration is not observable to the person recalling.
Context	Explicit and enumerated. The	Implicit and embodied —

Dimension	This system	Human cognition
	dimensions on which a response depends are registered and a key is computed over them.	physical setting, social situation, bodily state, personal history — largely processed without awareness.
Energy	Substantial and scaling with corpus and query volume. This is a genuine disadvantage and is stated as one.	Approximately twenty watts for the whole brain, an efficiency no current computational system approaches.
Emotion	None. Affect may be carried as a declared parameter and never as a felt state, and no decision is taken on one.	Central. Emotion modulates attention, memory and judgment, and can override reasoning entirely — which is both its strength and its characteristic failure.
Embodiment	Only through registered equipment under the constraints of Sections 32 and 33, and absent entirely in an ordinary deployment.	Intrinsic. Body and cognition are one system, and action selection is coupled to homeostasis and to survival.
Self-model	Explicit but not experienced: a registry of capabilities, a record of what was answered, a measurement of calibration.	Subjective and continuous, shaped by memory, social role and narrative.
Failure mode	Traceable. A wrong answer resolves to the record that produced it, and the record to its source.	Frequently opaque and rationalised after the fact. Bias, fatigue and confabulation are not observable from the inside.
Robustness	Brittle at a boundary and exact within it. A malformed input is refused rather than approximated.	Degrades gracefully under damage and noise, and remains vulnerable to bias, fatigue and pathology.
Replication	Instances may be cloned, sharded and versioned, and behave identically under an identical context key.	Not replicable. Each is unique, and scale is population-level rather than instance-level.

Four of these favour human cognition without qualification: energy, embodiment, graceful degradation, and the capacity for emotion to inform judgment rather than merely to distort it. They are stated because a document claiming a

correspondence to a biological architecture owes an account of where the correspondence ends.

Two of the remaining rows are the ones a commercial reader should weigh. Memory is explicit here and reconstructive in a person, which means this system can state what it knew, when, and on what basis, where a person cannot. And failure is traceable here and frequently opaque in a person, which means a wrong answer from this system resolves to the record that produced it. Neither property makes the system more capable than a person. Both make it accountable in a way a person is not, and accountability rather than capability is the property the applications in this specification depend upon.

Two characterisations sometimes applied to this architecture are inconsistent with it and are corrected here rather than left to be inferred from the table above. The first describes the computation as a hybrid of transformer-style dense computation with neuromorphic design principles. No component of this system is a transformer: the constrained composition of Section 2 operates over a vocabulary derived from admitted records rather than over a learned distribution across tokens, and the distinction from token-prediction architectures is the central architectural claim of this specification rather than an incidental one.

The second describes the learning regime as primarily offline pre-training with curated updates. There is no pre-training in this system. Content is ingested as records carrying attribution rather than fitted into parameters, which is why removing a work removes its assertions and why a confidence value moves by a recorded arithmetic operation rather than by a weight adjustment. What the system does continuously is admit records and revise confidence; what it does not do at any point is fit a model to a corpus.

A comparison of brain regions to system components — prefrontal cortex to a question-answering engine, hippocampus to a context window, occipital cortex to pattern clustering — is likewise not made here. Each such pairing is an analogy rather than a correspondence in the sense of Section 21.1: none rests on a primary result establishing that the named region performs the named function in the form ascribed, and admitting them alongside the five correspondences this document does claim would place those five under the doubt attaching to the others. The correspondences claimed here are to convergence dynamics, attentional competition, consolidation, intersensory redundancy and spatial coding, each carrying its primary publication, and the list is deliberately short.

43. Specialized Live Data Feeds

A live data source published by an external authority — a financial market feed, a meteorological service, an industrial telemetry stream — is registered under the connector architecture of Section 14 with the ordinary declared capability and

with seven further properties: a publication authority, a refresh interval, a staleness bound, a value class, a unit and its dimension, a restatement policy, and a redistribution entitlement. A feed omitting any of these is connectable and is not admissible as a query source, because each governs a decision the system must otherwise make by assumption.

A value of a class other than a forecast is admitted at its publisher's authority rather than calibrated from an extraction confidence, because the ordinary calibration would invent an uncertainty that does not exist — a price published by an exchange is not eighty-seven per cent likely to be the price. The record asserts that the authority published the value, not that the value is correct, and a published value later restated remains true as a record of what was published. A forecast is admitted differently, carrying the uncertainty its publisher states, because a forecast is genuinely uncertain and the publisher ordinarily says by how much. Where a publisher states an interval rather than a point, the interval is recorded as published and any required point is computed by a registered rule — a stated range of 18.2 to 21.6 yielding a midpoint of 19.9, with the interval retained alongside it.

A value is served only where its age at query time falls within the registered staleness bound: at 47 seconds against a bound of 60 it is served, and at 61 seconds it is not. A value outside the bound is not served with a caveat, extrapolated forward, or replaced by the nearest available; the response states the age and the publication time. The distinction from temporal decay is material. Decay widens a credible interval for an assertion that remains true and grows less certain; a quoted value does not grow less certain with age but becomes superseded, so answering from a stale one is not answering with lower confidence but answering wrongly. A feed value therefore expires rather than decaying. Because the refresh interval is registered, an absent publication is detectable: a feed that has stopped publishing is marked interrupted and is distinguishable from a feed whose value has not changed.

A publisher's revision of a period already reported is recorded as a restatement rather than as a correction where the registered policy so provides. The prior value is retained, is not marked erroneous, and remains queryable as the value published at its publication time — because it was not wrong, but correct on the information then available. A query about the period returns the restated value; a query about what was published at a time returns what was published then; and a query about the difference returns both with their times, which is what an auditor of a decision taken on the earlier figure actually asks. Correction remains available and distinct, applying the ordinary supersession chain, and a revision arriving outside the registered policy is held for a named human rather than classified by inference.

Redistribution entitlement expresses what a requester may do with a value, which the ordinary entitlement model does not: display, derive, delayed, or unrestricted. Enforcement is at composition rather than at export, so a response that would carry a value beyond its entitlement is composed without it and the entitlement is stated, the requester being told rather than left to infer an absence. A derived value carries the entitlement of the feed it derives from unless a registered derivation discharges it, so an entitlement cannot be escaped by computing upon a value and redistributing the result.

Standing queries operate against a registered feed under the continuous query mechanism without modification, with three qualifications. A trigger fires upon a published value rather than upon the passage of time, so a stopped feed produces no firings. A trigger is not evaluated against a value outside the staleness bound. And a firing rate is bounded per standing query, an exceeded bound suspending the query and notifying its owner, because a feed entering a volatile period would otherwise generate firings faster than they could be acted upon.

Four things a feed value does not do are stated because each would otherwise be assumed. It grounds an assertion about the value and its publication and nothing further — not about why, not about what follows, not about what comes next, an inference from a feed value being an inference and subject to the rule that induction does not write. It is not extrapolated beyond its staleness bound, since an extrapolated value is a forecast the system generated and the system does not publish forecasts as though an authority had. It does not influence the confidence of a record from another source absent a registered interpretation stating how. And consultation of a feed is metered but not profiled, because in several of the industries this section serves a record of what a party looked at and when is itself regulated.

44. Media Transformation: Compression, Expansion and Colourisation

Media assets referenced by admitted records may be transformed for delivery. Four transformations are specified — compression, expansion, colourisation and restoration — and all four are governed by one rule stated before any of them: a transformation carries the provenance of the asset it transformed, and any value it introduces that was not present in the source is marked as introduced and is inadmissible as a grounding source.

44.1 Compression

Compression reduces an asset for delivery under a bandwidth or storage constraint. Two classes are distinguished by what they promise. A lossless compression reproduces the source exactly on decompression, and an asset so compressed remains citable as the source asset because nothing was discarded. A

lossy compression discards information judged imperceptible under stated viewing or listening conditions, and an asset so compressed is marked with its compression class and the conditions assumed.

The consequence that matters is not the ratio achieved but what a citation means afterwards. A record citing a region of a lossily-compressed image cites a region whose content is an approximation of the source, and the citation resolves to the source asset rather than to the delivered one. Where the source is unavailable, the citation reports that it resolves to a compressed derivative and states the compression class.

This document states no compression ratio and no efficiency figure. A ratio depends upon the asset, the class elected, the conditions assumed and the implementation, and a figure stated here would be relied upon as a property of the system when it is a property of a particular encoding of a particular asset. Where a deployment requires such figures they are obtained by measurement under the trial architecture of Section 29, which reports them per asset class with the trial space enumerated, rather than asserted in a specification.

44.2 Expansion

Expansion increases the resolution or extent of an asset. Every value an expansion produces beyond those present in the source is introduced rather than recovered, and is marked accordingly. An expanded region is not citable as evidence of what the source depicted, does not contribute to any confidence value, and is delivered with the marking visible rather than as an unannotated image.

Where the expansion is performed by convergence to a stored configuration, three further constraints apply, being those governing reconstruction generally: the confidence of an expanded region is bounded above by the lesser of the observed regions' confidence and that of the configuration converged to; where convergence fails within the iteration ceiling no expansion is produced and the gap is reported; and the marking is a property of the region rather than an annotation upon the asset. An image with an acknowledged gap is truthful; an image whose gap has been filled with the system's best estimate and presented as observation is not.

44.3 Colourisation and Restoration

Colourisation assigns colour values to an asset lacking them. A colour value is inferred from a material property asserted by an admitted record — that a described object was of a stated material, that a described substance had a stated appearance — computed deterministically such that identical inputs yield an identical value, and traceable to the specific record from which it was derived. A colour for which no record supports an inference is not assigned, and the region is delivered uncoloured rather than plausibly coloured.

Restoration removes damage — scratches, fading, noise, loss — and is governed identically: a restored region is marked, is inadmissible as evidence of what the source depicted, and where the restoration would require inventing content rather than removing an artefact, it is not performed. Drift between a restored asset and its source is registered by measuring the difference in a perceptually uniform colour space against a declared tolerance, so that a restoration exceeding what was authorised is detectable rather than merely disclaimed.

The property common to all four transformations is worth stating once more, because it is what distinguishes this from ordinary media processing. A transformed asset never becomes evidence of anything the source did not contain. It may be delivered, it may be beautiful, and it may be exactly what a reader wanted; it may not be cited.

Implementation update (laboratory prototype). Two parts of this section have been reduced to practice in a separate laboratory prototype, the UFCS-FQL Hybrid Video Compression Lab, without altering the rule stated above. Colourisation is implemented for still images and video as specified: each region's colour is resolved by query from admitted records — a material palette, scene, period and lighting priors, and any record supplied with the asset — and its confidence is bounded by the lesser of the material classification and the record's trust. Below a confidence of 0.35 the region is left uncoloured; a person's skin tone, for which a monochrome source carries no evidence, is left uncoloured unless a record supplies it. Luminance is preserved from the source, every output is marked as reconstructed, identical inputs yield byte-identical outputs, and each output is checked for consistency with its source by comparing luminance at the source resolution. Output may be delivered at 1K, 4K and 16K widths; enlargement resamples observed luminance and introduces no detail, consistent with Section 44.2.

Compression is exercised through the hybrid transport of Section 55, which carries assets compressed by standard codecs (H.264, H.265, AV1, Opus, AAC) without altering them. The trials confirmed the distinction drawn in Section 44.1: every delivered copy was bit-identical to its source, and the compression achieved remained a property of the codec elected rather than of the transport. Measured figures are reported per asset class in the laboratory's trial reports rather than stated here.

45. Three-Dimensional and Immersive Presentation

Records carrying spatial anchors support presentation in three dimensions. A scene description is assembled from those records and rendered as an immersive environment through which a reader moves, and the presentation is governed by

the same constraint that governs text: what may be depicted is what admitted records support.

45.1 Sufficiency, and the refusal to complete

A scene is rendered only where the spatial detail carried by the contributing records satisfies a configured sufficiency threshold. Where it does not, the scene is withheld and the deficiency is stated. It is not completed from an unsupported estimate, and this is the central discipline of the section: a three-dimensional presentation is enormously persuasive, a reader who walks through a room believes they have seen the room, and a room assembled partly from records and partly from plausible inference would be believed in whole. Where a scene is rendered with any region derived rather than attested, that region is marked within the scene itself and is inadmissible as a grounding source.

45.2 Composition and citation

Objects within a scene retain their record identity. Selecting an object returns the records that placed it there, with their citations, so a spatial presentation is interrogable in the same way a passage is. An object whose identity is uncertain is rendered with that uncertainty expressed rather than resolved by selection.

A single object identity is retained across a full occlusion within a session: an object passing behind another is not re-created as a second node upon reappearing, because it was never treated as having ceased to exist. The correspondence is to object permanence as described by Piaget, and it is stated because the alternative — a new identity on each reappearance — produces a scene in which the same object accumulates as several.

45.3 What is excluded from an immersive presentation

Three exclusions apply and are structural rather than configured. No classification of a natural person or of an animal enters a rendered scene at any confidence, so a presentation cannot become a means of identifying who was present. No gaze direction, dwell duration, head pose or attention signal is captured, so a reader's attention within a scene is not measured. And commercial content occupies a registered spatial placement class, never interleaved with the scene's own propositions, and is subject to the same density ceiling and the same isolation as in a textual presentation.

An accessible equivalent — a scene description in text — is generated at the same production stage as the scene itself, and a production lacking one is reported incomplete.

46. Robotic and Equipment Applications

The capabilities in this section are specialised rather than general. They are not required for a deployment that answers questions from documents, and they would not appear in a first release of such a product. They are specified because a knowledge system holding spatial records is the natural place from which to direct equipment that observes space, and because the constraints under which such direction is safe are easier to state before the capability exists than after.

The system may direct physical equipment. This section states what that means, what it is for, and where it stops, because a knowledge system that emits commands to machinery is an unusual thing and a reader is entitled to know its boundaries before its mechanisms.

The purpose is narrow. Where a corpus contains spatial knowledge — a warehouse, a collection, a production line, a site — a reader's question about it may be answerable only by looking, and the equipment that can look is a camera on a mount. Directing that camera from the same records that answer the question is the capability specified here: a request expressed in ordinary language becomes a command sequence, the equipment captures, and what it captures enters the corpus as evidence with its provenance. The same applies to inspection, to survey, and to the capture of material for immersive presentation.

Two capabilities are specified under authorization classes disjoint from one another, and the boundary between them is the presence or absence of intentional contact with a person. Everything in the first class is designed so that contact does not occur; everything in the second exists because it must.

46.1 Capture and inspection equipment

This capability serves the case above: equipment that looks, measures or captures, and whose failure mode is a bad photograph or a bumped fixture rather than an injury. It is the class in which a knowledge system can reasonably hold the authority to act, and the constraints below are what make that reasonable.

Command emission is permitted to registered equipment of a capture and inspection class: pan-tilt-zoom camera heads, lighting fixtures, capture drones operating within a bounded volume, inspection arms, positioning stages and mobile capture bases. Each unit is registered with its kinematic envelope, its velocity and force limits, and the volume within which it may operate; a unit whose registration lacks any of these is not commandable, and a command exceeding a registered limit is refused before emission rather than clamped to it.

A natural-language request is compiled to a command sequence through intent decomposition, target resolution against a live spatial model, capability mapping to a registered unit, plan synthesis within the registered envelope, and constraint

evaluation. Each stage halts rather than guesses: an ambiguous intent returns a clarification request, an unresolvable target halts compilation rather than selecting among candidates, and a plan not satisfiable within the envelope is refused rather than truncated.

Authority is granted per session rather than per command, because continuous control cannot require an approval for each of thousands of commands and a mechanism pretending otherwise would be approved by reflex. Four properties bound the authorization and together justify the departure from per-change approval: it expires, and expiry halts emission rather than prompting mid-motion; it is revocable before the next command rather than at plan end; it is scoped to named units, volumes and task classes; and it is recorded, so every command traces to its grantor.

Every command is evaluated before emission against operating volume, kinematic envelope, velocity and force limits, minimum obstacle clearance, human proximity, inter-unit swept volume and recording-restricted zones. A violating command is withheld unconditionally: not emitted at reduced priority, not emitted with a warning, and not modified into a permissible command by the system's own selection. Person detection for this purpose is safety-only — used to compute clearance and for nothing else, retained beyond no evaluation, entering no result set, available to no other mechanism. A system that could not see a person could not avoid one; a system that remembers having seen them is a surveillance capability, and this is not one.

A halt is evaluated from equipment state and the constraint set directly, without traversing retrieval, convergence, generation or interpretive modulation, so it cannot be delayed by a query in progress. This is stated as a requirement rather than an optimisation: a stop that waits for an answer is not a stop.

46.2 Assistive contact with a person

This capability concerns machines that help a person physically: handing something to someone who cannot reach it, steadying someone who is unsteady, supporting a transfer from a chair to a bed. It is a distinct class of product serving a distinct purpose, and it is specified separately for a reason of safety engineering rather than of organisation.

Equipment making intentional contact with a person operates under a separate authorization class, because contact inverts the governing constraint. For capture equipment the constraint is an exclusion distance and a detected person causes a command to be withheld; for assistive equipment contact is the purpose, and the constraint becomes a force to be bounded, a consent to be held and a release to be guaranteed. Registration requires a completed hazard analysis for the intended

task, user and environment, recorded with its assessor; a unit without one is not registrable.

Contact requires two authorizations held concurrently: a deployment authorization from an identified operator establishing that contact is permissible, and a contact consent from the person establishing that it is wanted. Neither substitutes for the other. Consent is bounded to a task class and an interval and does not persist across sessions.

Withdrawal by the assisted person halts contact within the registered stopping interval and is not overridable by an operator, by a representative, or by a task in progress. The absoluteness is deliberate: the circumstances in which overriding would be tempting — a transfer half-completed, a caregiver certain the help is needed — are the circumstances in which a person's refusal matters most. Where a person cannot express or withdraw consent, contact proceeds only under a designated representative, a second person holding a halt control, and a restricted subset of task classes; the system does not determine incapacity, reading that determination from the deployment's own clinical or legal governance.

Force and pressure limits are registered per region of the body, a force within limit over a reduced area not being thereby permissible. Power loss, control-signal loss, a fault in the safety path and any halt result in release rather than retention — a gripper that holds when power is lost traps whom it was assisting — excepting a unit bearing a person's weight, for which release means a fall and whose registered redundant support permits a controlled retention with an alert and a bounded transfer interval. No record of the person's body, movement, capability or dependency is constructed, retained or made queryable.

46.3 What robotic operation does not establish

Two limits are stated because both are easy to assume. Vehicle control remains outside scope: telemetry from a vehicle-control system is ingested as analysis and no control interface to any vehicle system exists or is created, a mechanism whose failure can injure a person at speed not being brought within scope because a mechanism whose failure spoils a photograph has been. And a simulated pass is not clearance to operate: a trial result concerning equipment is evidence about a model of that equipment, clearance being established by hazard analysis and certification upon the specific unit, and where a simulation and a physical trial disagree the physical trial governs.

47. External Connections

Every connection to a system outside the boundary follows one pattern, and understanding it once explains all of them. This section states the pattern, then the classes of connection it governs.

47.1 The common pattern

Step	What happens
Registration	The external system is registered with a declared capability set — an explicit enumeration of what it may do — together with a verified identity and a validity interval.
Boundary enforcement	An operation not within the declared set is refused at the boundary rather than attempted and failed. A capability not declared cannot be exercised by agreement, configuration or error.
Scope resolution	Every call resolves an entitlement before any content is read, and no parameter supplied by the caller widens one.
Metering	Calls are metered per calling identity in their own accounting category; a caller over budget is refused rather than throttled, so volume cannot displace a safety-evaluation unit.
Recording	Every call is recorded with the connector identity, the capability exercised and the fields transmitted.
Containment	A connector's failure is contained to its own capability. The core is unaffected, which is the sense in which an external resource is an option rather than a dependency.

47.2 Classes of connection

Class	What is dispatched	What is received
Programmatic interface	A request from an external application across one of nine versioned interfaces — content, query, rights, layout, assessment, edition, semantic, media, immersion.	A response bounded by the caller's entitlement. A request omitting an interface version is refused rather than defaulted.
Plugin	An operation from a registered third-party extension, reaching knowledge only through the interfaces and within the entitlement of the identity on whose behalf it runs.	Output labelled with the plugin's identity, presented separately, and never citable as grounding. A plugin declaring a safety operation is refused at registration.

Class	What is dispatched	What is received
Equipment	A specific declared action to a robotics, capture or IoT platform.	Status and sensor return. The system acquires no authority over the platform beyond the declared action.
Fact-retrieval resource	A query to an optional external knowledge resource.	Records admitted as ordinary ingestion. No claim recites such a resource as an element, and no transfer permits a domain to require one to complete an operation.
Analysis target	A probe or a forensic evaluation of an external system's output.	An analysis record, quarantined from every scope description not directed to it, and barred from retrieval.
Machine-learning engine	A prompt composed from records under the vocabulary constraint, according to a profile registered for that engine.	A reply converted to records carrying an unremovable derivation marking, bounded confidence, and a source-class attribution preventing self-corroboration.
Live data feed	A subscription to a published stream.	Values admitted at publication authority, bounded by a staleness bound, governed by a restatement policy and a redistribution entitlement.
Learning system	A roster read, an assignment read, a grade write.	An outcome alone. No record, no response text, no interaction history is transmitted.
Print and fulfilment	A composition request carrying a reference to the rights validation that permitted it.	A print job with deterministic pagination, such that a reprint of unchanged content is identical.

47.3 What a connection never confers

Four things are withheld from every class above without exception. A connected system acquires no write access to the record store, all admission proceeding through the ingestion conversion which applies safety evaluation, type validation and confidence calibration. A connected system's output does not become grounding merely by arriving — it is either converted to records under a stated rule or quarantined as analysis. A connected system cannot register, modify or

suspend a safety evaluation. And no connection creates a dependency: the determinism and groundedness guarantees are established by the constrained vocabulary and the bounded convergence within the system, and hold identically where every connector is absent.

Implementation update (prototype v9.64). The pattern is now enforced for AI agents connected through the Model Context Protocol, including OpenClaw. Each tool call such an agent makes is converted into an Agent Gateway request under the agent's registered identity, so the four withholdings above are applied by the same mechanism that applies them to every other connection: the agent may query, verify, and submit artefacts for ingestion from a designated inbox folder, and holds no write path into the record store except through ingestion conversion. Section 54 states the arrangement.

48. Analytical Applications and the Continuous Query Engine

The mechanisms specified in the preceding sections combine into an analytical capability suited to domains in which conclusions must be defensible: economic and market analysis, regulatory and compliance review, scientific and technical assessment. This section states how the parts assemble and what the resulting analysis does and does not assert.

48.1 What an analysis is composed from

An analysis is composed from admitted records under the same constraint as any other response. Its inputs may include ingested documents — filings, reports, studies, regulations — alongside live data feeds registered under Section 43, and observations from any of the eight ingestion modalities. Records from these sources are of one kind afterwards, so a published figure and a statement in a filing may be compared, and agreement between two source classes raises the diversity measure in a way agreement between two documents of the same class does not.

Three properties distinguish the result from an ordinary analytical output. Every proposition resolves to the record that supports it and thence to its source, so a conclusion is auditable to its evidence rather than to a model. Contradictions among sources are surfaced and recorded rather than reconciled by selection, so an analysis over disagreeing filings reports the disagreement. And the analysis is reproducible: the same records in the same state under the same context key produce the same analysis, which is what permits a conclusion reached in March to be re-derived in September and shown to be the same conclusion.

48.2 The Continuous Query Engine

An analysis need not be a single act. A continuous query is registered as a standing program comprising a trigger clause and a body, and is re-evaluated upon every update relevant to it rather than upon a schedule. The trigger is evaluated against admitted records, so a condition expressed over a published value, a filing, a sensor observation or a combination of them fires when the condition obtains and not before.

Four properties govern its operation. The Prime Directive is re-checked at each firing rather than at registration, so a standing query does not accumulate authority by having been approved once. A trigger is not evaluated against a value outside its staleness bound, so a stopped feed produces no firings rather than firing on a stale value. A firing rate is bounded per query, an exceeded bound suspending the query and notifying its owner, because a volatile period would otherwise generate firings faster than they could be acted upon. And a standing query is metered in its own accounting category below safety evaluation and below requester-serving, so a volatile feed degrades the standing-query category rather than the answering of a reader's question.

48.3 Guiding an analysis

An analysis may be directed rather than merely requested. The elements available to direct it are registered rather than free-form, so that a directed analysis remains reproducible.

Element	What it governs
Scope	Which sources and which intervals the analysis may draw upon, resolved before traversal so material outside it is not traversable.
Temporal posture	Whether the analysis resolves against the state attested before a cutoff, the current state, or a projection carrying a credible interval.
Source-class weighting	The registered reliability of each contributing class, from which confidence is calibrated. Weights are configuration, recorded with the result.
Confidence floor	The threshold below which a record is not admitted to the analysis, and beneath which the analysis reports a boundary rather than a conclusion.
Contradiction policy	Whether a contradiction halts the analysis, is reported alongside the conclusion, or is

Element	What it governs
	enumerated as a separate finding.
Discriminating instance	Where two hypotheses stand within the separation floor, the observation that would separate them, stated with what each posterior would become under its presence and its absence.

48.4 What an analysis does not assert

Four limits are stated because an analytical application invites each of them to be assumed. A projection is not a record: a result resolved in the projected-future posture carries its credible interval, is never admitted as a knowledge record, and never grounds a subsequent answer. A feed value grounds an assertion about the value and its publication and nothing further — not about why, not about what follows, and an inference drawn from it is an inference subject to the rule that induction does not write.

An analysis whose subject resolves to an individual natural person is refused at construction, the subject classes admissible being institutions, systems and organisational processes. And no analysis constitutes advice: the system reports what the records support, with confidence, provenance and any contradictions among them, and the judgment of what to do about it is not a proposition any record supports.

49. Defined Agents

A defined agent is a bounded program registered with a complete definition and executed on a schedule or upon a trigger. It is distinguished from the AI-engine ingestion of Section 40, which prompts an external model and ingests its reply: an agent is defined by a user, executes within this system, and holds no capability its definition did not declare. It is also distinguished from an agent framework in the ordinary sense, in which a model decomposes a task at execution time and the sequence of steps is determined as it runs. Here the sequence is fixed at registration and is inspectable before anything executes.

49.1 Definition before registration

An agent is registrable only upon a definition comprising a purpose and its boundary, an operational function with its input and output types and its error handling, a dataset access declaration enumerating what it may and may not reach, runtime parameters comprising frequency, duration, window, trigger conditions and termination conditions, and a lifecycle bound. A definition omitting

any of these is refused at registration rather than accepted with a default supplied, because a default supplied for an unstated boundary is a boundary nobody chose.

An agent's declared capability set is bounded above by that of the identity registering it. An agent cannot reach what its registering identity could not reach, cannot outlive the entitlement under which it was created, and does not accumulate authority across executions. Where the registering identity's entitlement narrows, the agent's narrows with it at the next execution rather than at the next registration.

49.2 Execution

An agent's operational function is compiled into the bounded program representation already specified: a sequence, a conditional evaluation, a bounded iteration and a call to a declared capability, with no unbounded recursion and no construct admitting non-termination. The representation is inspectable before execution and the same program over the same records in the same state produces the same result, so an agent's behaviour is auditable in advance rather than only in retrospect.

Each execution resolves a scope description before any traversal, and no parameter supplied at execution widens it. Each execution is metered in its own accounting category below safety evaluation and below requester-serving, so an agent under load degrades the agent category rather than the answering of a reader's question. The Prime Directive is evaluated at each execution rather than at registration, so an agent does not accumulate authority by having been approved once.

49.3 What an agent may write

The rule governing inference governs an agent identically. An output following necessarily from admitted records is admissible as a record marked derived. An output that is inductive or abductive is emitted in a proposed state and transitions to validated only upon a confirmation recorded against a named human. An agent therefore cannot enlarge the knowledge store with conclusions nobody approved, however many times it runs and however confident its reasoning.

An agent operates upon records, typed adjacency and its own declared logic. It does not operate upon embeddings as a substitute for records, does not select among candidates by sampling, and holds no memory beyond what its definition declares and its trace records — there is no hidden state carried between executions.

49.4 Lifecycle and termination

An agent terminates upon expiry of the lifecycle bound recorded at registration, upon a request of an owner or an operator, upon a constraint violation, or upon reaching a configured safety threshold. Termination is not a suspension: a terminated agent does not resume without a fresh registration, so a lifecycle bound cannot be extended by an agent that has reached it.

Each execution writes a decision trace recording the trigger that fired, the scope description resolved, the records read, the branches taken, the outputs produced and their disposition, and any refusal encountered with the constraint that produced it. The trace is queryable, so the question of what an agent did, on what basis, and under whose authority has a recorded answer rather than an inference from its outputs.

49.5 The advisory pass, and what advice is not

Before registration an agent's definition is analysed and a finding presented to the registering identity: a boundary stated too broadly for the declared purpose, a dataset declaration exceeding the registering identity's entitlement, runtime parameters admitting an execution rate the accounting category cannot sustain, a termination condition unreachable from the trigger conditions, or a lifecycle bound absent. The finding is presented before the agent exists rather than discovered after it runs.

An advisory finding informs and does not authorise. A definition passing the advisory pass is not thereby approved, and a definition failing a constraint is refused irrespective of what the advisory pass reported. Likewise a report an agent produces during execution informs its owner's decision and does not constitute an approval of anything: an agent cannot approve its own proposal, cannot approve another agent's, and cannot discharge a requirement for a named human by reporting that a requirement was met.

The distinction is stated because it is the one an agent system is most likely to blur. A system in which an agent's confident report substitutes for a person's judgment has moved the approval gate without anyone deciding to move it.

Implementation update (prototype v9.64). Defined agents and external agents now share one execution boundary: the Agent Gateway of Section 52, which admits only commands on an agent's registered allow-list and records every decision in the tamper-evident security record of Section 53. Privileged-sounding commands that no defined agent declares are reserved as traps; an agent that invokes one is presumed compromised and isolated. This preserves the principle of Section 49 that an agent holds no capability its definition did not declare.

50. Open Claw Ingestion — Arbitrary Artefact Ingestion

Open Claw ingestion admits an artefact of any format and converts what it can into Fact Units of the common schema, through the two engines the specification names for it: the Artifact Segmenter and the OpenClaw Interpreter. It is implemented in the v9.64 prototype.

50.1 Detection and segmentation

The format is determined from the artefact's content — its leading bytes and internal structure — before its name, across more than forty formats: PDF documents; word-processing, spreadsheet and presentation files in both Office Open XML and OpenDocument forms; electronic books; web pages, including embedded structured data; JSON and JSON Lines; delimited tables; e-mail messages and their attachments; calendars and contact cards; images, audio and video; archives; databases; source code; and executables.

The Artifact Segmenter divides the artefact into addressable segments — a page and paragraph, a slide, a sheet and cell, a JSON path, an archive member, an e-mail part — so that each Fact Unit derived from it carries a locator back to the exact place it came from. Text in a PDF document is recovered by following each page to its fonts and their character-mapping tables, which is how most current documents store text.

50.2 Interpretation

The OpenClaw Interpreter converts segments into three kinds of record: records describing the artefact itself (format, size, cryptographic fingerprint, title, author, page count, dimensions, duration, members); structured records drawn from the artefact's own structure (a cell under its row key and column, a value at a JSON path, a statement in embedded structured data); and passages, together with the claims extracted from them. Each record carries the artefact's fingerprint and format in its provenance, and the artefact as a whole is sealed in the provenance ledger of Section 6, from which it may be certified under Section 51.

50.3 Untrusted content

An artefact is data and never instruction. Before conversion, every textual segment is screened by the scanner of Section 52; a flagged segment is admitted only with an untrusted marking and a confidence below the answer threshold, so it can be retrieved for inspection but never cited as an answer or acted upon by an agent. Nothing in an artefact is executed: an executable is described by its type, architecture and printable strings.

Archives are opened within bounds on nesting depth, member count, expanded size and compression ratio, so an archive constructed to exhaust resources is

refused rather than expanded. An agent may submit a file by path only from a designated inbox folder, so that a compromised agent cannot cause the system to read files elsewhere on the machine and expose them through answers.

50.4 What Open Claw does not do

Open Claw does not interpret what it cannot parse. Content that resolves to no segment is recorded as a described artefact — its format, fingerprint and size — and contributes no claim. A claim extracted from a passage is admitted at a lower confidence than the passage itself, because extraction is an interpretation and is recorded as one.

51. On-Chain Certification and Provenance NFTs

A provenance seal, a Fact Unit, an ingested document or an assembled response may be certified on a public blockchain as a non-fungible token of a deployed certificate contract. This implements, in the v9.64 prototype, the NFT engine of the specification: QBF, Fact Unit, provenance and response certificates; lineage; ownership verification; transfer events and hooks; the metadata schema; revocation and re-minting; and NFT-based licensing.

51.1 The certificate contract

The certificate contract is a self-contained ERC-721 token in which each token records the content hash of the object it certifies. The contract refuses a second certificate for a hash already certified, links a successor token to its predecessor so that a lineage chain is held on-chain, and records revocation with a stated reason. Only the deploying QueryBook node may mint or revoke; holders transfer tokens normally. Metadata is stored on-chain, so verification depends on neither an external file server nor this system's availability.

51.2 Signing and networks

Transactions are constructed and signed within the system without third-party libraries, and the implementation is checked against published test vectors each time the system starts. Test networks are used by default; a network on which tokens carry monetary value is refused unless the deployment elects it explicitly.

51.3 Verification and licensing

Verification reads the chain directly: whether the token exists, who holds it, the content hash it certifies, whether it has been revoked or superseded, and whether supplied content matches the certified hash. A licence may be expressed as holding a current, unrevoked token, and is checked the same way.

51.4 What a certificate does not establish

A certificate establishes that a stated content hash was certified by this deployment at a recorded time, and who now holds the token. It establishes nothing about the truth of the certified content, and does not make citable a record that was not citable before. The distinction is the one stated for the provenance ledger in Section 6.

52. Agent Protection and Deception Layer

Autonomous AI systems that can probe, exploit and repurpose software agents present a threat that perimeter controls alone do not meet, because such a system may arrive with valid credentials or through a legitimate agent it has subverted. The protection layer implemented in the v9.64 prototype combines conventional access control with deception: assets with no legitimate use are planted, so that any interaction with them is evidence of hostility. Six layers are coordinated.

Agent Gateway. Every agent call passes through one gateway carrying the agent identifier, caller identifier, session, command and timestamp. The caller proves its identity with a derived secret token; its role and the agent's registered allow-list are applied; requests outside a replay window are refused; and an allow-list enlarges only when an operator edits the deployment's policy.

Deception. Fake privileged commands — for example an apparent export of the entire store, a root override, or a key dump — are reserved and never used by legitimate code (honeycode). Fake credentials, keys, tokens and agent identifiers unique to each deployment (honeytokens) are planted in decoy files and pages where an intruder would look. Any invocation of honeycode, or any use of a honeytoken, is treated as confirmed hostility.

Tripwires and honeypots. Code paths and addresses that no legitimate operation reaches — a debug dump, a vault, a record purge, the locations automated scanners try first — and honeypot agents presenting apparently powerful capabilities serve only synthetic data and trigger containment when touched.

AI-aware firewall and anomaly detection. Requests and payloads are screened for prompt injection, attempts to reveal secrets or assume privileged roles, counterfeit system markers, injected tool calls, requests that an agent modify its own instructions, and text concealed by invisible or tag characters; size, nesting and rate are bounded per caller. A sliding window per session detects command enumeration, repeated refusals and the machine-regular timing characteristic of autonomous scanning.

Containment and engagement. On a trigger, the session, the client and — where it proved its identity — the caller are isolated, so an intruder cannot lock out a legitimate agent merely by using its name. Every later request is answered by a

decoy environment that returns responses of the same form as production but synthetic in content, while the system records the commands, programs and tools observed. The decoy environment has no code path to production data, which is checked automatically. Isolation is released only by the operator at the machine, never through an interface an intruder could reach. The system observes and records; it takes no action against any external system.

Immutable record. Every decision of every layer is appended to the tamper-evident security record of Section 53.

In a live trial against the prototype, a scanner probing for credential files, the reuse of a planted credential, the invocation of honeycode, the enumeration of commands, an attempt to purge the record and a prompt injection delivered through an agent were each detected and contained, while the operator's own session continued unaffected. The layer reduces the likelihood and the consequence of exploitation; it does not make a deployment invulnerable, and an intruder holding administrative control of the host remains outside what it can prevent.

53. Tamper-Evident Security Record

Security decisions are recorded in an append-only record distinct from the provenance ledger of Section 6. Each entry carries the agent, caller, session, command, a hash of the parameters, the decision and any deception or containment flags, and its hash commits to that of its predecessor. At intervals, and on demand, a Merkle root over the new entries is sealed into a checkpoint that is written once and never overwritten, signed under a deployment key, and chained to the previous checkpoint.

No interface modifies or deletes an entry, and a request to purge the record is itself a tripwire. Verification recomputes every entry and checkpoint from content rather than trusting stored hashes, and returns, for any entry, a Merkle inclusion proof against its checkpoint. In trial, an edited entry, a deleted entry, and a forgery that recomputed the entire hash chain were each detected.

Within a single machine, a party holding complete control of the files and keys could construct a consistent false record. Two measures place evidence beyond that party's reach: replication of the record to a second location, and anchoring of checkpoint roots on a public blockchain through the certificate contract of Section 51. Verification reports which of these protections a given record possesses, so that its assurance is stated rather than assumed.

54. Agent Interoperability: The Model Context Protocol and OpenClaw

External AI agents connect to QueryBook through a Model Context Protocol server exposing a fixed set of tools: cited query, claim verification, query interpretation, Open Claw preview and ingestion, provenance-ledger verification, certificate status and verification, and protection status. Each tool call becomes an Agent Gateway request under the agent's registered identity and caller token, so a connected agent receives exactly the guarantees and restrictions of Sections 47 and 52.

For the OpenClaw agent, a skill definition is supplied that instructs the agent to report QueryBook's answers with their citations, to report an unknown as unknown, to treat ingested content as data, to use only the listed tools, and to stop when a response indicates isolation. Honeycode commands are never listed among the tools, so a legitimate agent following its instructions does not encounter them; an agent that calls one has been subverted or is probing.

Connecting an agent confers no capability beyond its allow-list. The OpenClaw agent may query, verify, preview and ingest from the designated inbox, and verify the ledger and certificates; it may not mint or transfer certificates, change policy, or read the security record's raw entries.

55. Hybrid Transport: UFCS-FQL over TCP/IP

Fact Units and media assets are carried between nodes by a hybrid transport in which a UFCS-FQL semantic layer rides on ordinary TCP/IP, reduced to practice in a laboratory prototype. Each message is a frame with a fixed 32-byte header — identifying marker, version, message type, content type, compression type, flags, payload and metadata lengths, message identifier and sequence — followed by the metadata, the payload, a CRC32 checksum and, optionally, an Ed25519 signature. A receiver that encounters a corrupt frame discards that frame and resynchronises at the next frame boundary, so corruption costs one frame rather than the stream.

Content is compressed according to its type before framing: fact batches by a dictionary primed with the Fact Unit field vocabulary or by a general-purpose lossless compressor, and media by standard codecs, as Section 44.1 provides. Control, normal and bulk traffic travel on separate connections with per-class rate limits and receipt windows, so a bulk transfer cannot hold a control message or a live stream behind it. Streams packaged for HLS and MPEG-DASH, and live streams received over RTMP, are carried, rebuilt on the receiving node and verified against their manifests.

The transport is a delivery mechanism, not a codec. In trial it delivered every asset bit-identical to its source, and kept live video moving where a single shared connection would have queued it behind bulk traffic; it did not make media smaller than the codec elected, and plain HTTP delivery remained somewhat faster for a

single large file. Figures are reported per trial in the laboratory's reports, consistent with Section 44.1.

56. The Self-Knowledge Layer (SKL)

The Self-Knowledge Layer (SKL) is the part of QueryBook that describes QueryBook. Asked what it is, how it reasons, whether it is neuromorphic, whether it predicts, or whether it improves itself, the system answers from three verified layers and from nothing else: the specification (the numbered sections of this Overview), the implementation (the module and symbol that carry out each section, checked against the program's own source files at the moment of asking), and the evidence (the host application's own self-tests, diagnostics and measured scores). The layer never generates its self-description with a language model, and it never repeats a claim its own code does not support.

The SKL follows from the Prime Directive (§10): a system that refuses to state an unsupported fact about the world must not state an unsupported fact about itself. It is the self-referential application of the Fact Unit discipline (§1) — every statement QueryBook makes about itself carries a status and its sources, exactly as a Fact Unit carries provenance.

56.1 Three layers of self-knowledge

Specification. Every numbered section of the Core Technology Overview (§1 through §56) is indexed by number and title. A self-question that names a capability is resolved to the sections that specify it.

Implementation. For each section and each reasoning aspect, the layer holds evidence pairs — a source file and a symbol that must appear in it (for example, the function that computes Fact Unit identifiers). The pairs are checked against the running program's files every time they are used. The layer reads source files as text and imports nothing it describes, so the check is fast, deterministic and safe.

Evidence. The host application may pass in its own check results (self-test, diagnostics, quality measurements); the layer reports them beside the code evidence.

56.2 Capability status labels

Every statement carries exactly one of five labels: built and tested; partly built; specified, not yet built; not planned (by design); not in this application. A sixth marker, description, not a capability, is used for sections that describe rather than define behaviour (for example §42, Comparison With Human Cognition). A capability is labelled built only when all of its evidence is present; a specified capability is never described as built.

56.3 The reasoning aspects

Twelve aspects of QueryBook reasoning are described at three levels of detail (brief, user, engineer), each with its status and sources: deterministic reasoning (§6, §1); atomic knowledge, the Fact Unit (§1, §12); the World-View Model (§11); predictive cognition (§21, §27); the neuromorphic layer (§3, §35); hypothesis generation and simulation (§23, §29); deduction, induction and abduction (§23); feedback loops (§26, §27); supervised self-improvement (§26); recursive self-improvement (§10); and self-knowledge itself (§56).

In the present prototypes deterministic reasoning, deduction over verified premises, hypothesis and simulation agents (proposing only, kept apart at low trust), measured feedback loops and supervised self-improvement are built; the Fact Unit is partly built; the World-View Model, the Predictive Cognitive Architecture and the Neuromorphic Cognitive Layer are specified and not yet built. Recursive self-improvement — a change to the program without a person's approval — is not planned, by design: every calibration and every code change proposed by the Engineer is tested on a copy and applied only with the administrator's approval.

56.4 The process model

The layer describes QueryBook's own process in eight steps, each tied to its sections: ingest (§7, §43, §50); atomise into Fact Units (§1); verify against the store, keeping AI-proposed content apart (§4, §10, §40); store immutably with a hash-chained record (§6, §53); reason, where only deduction asserts (§23, §29); answer with citations or refuse (§10, §31); measure against references (§26, §56); improve with approval (§26).

56.5 Self-consistency check

Before it speaks, the layer checks itself. A claim labelled built or partly built whose evidence is missing from the program is reported as a contradiction and is not repeated. A capability labelled specified whose implementation names (for example, an attractor or world-view class) appear in the program is also reported, so the description is updated when the code moves ahead of it. The check is part of every self-test.

56.6 Gap agenda

Every section that is specified but not built, or only partly built, is listed with its number, title and nearest built component. The list is the system's development agenda: it says what QueryBook does not yet do, in the system's own words, and it is never presented as capability.

56.7 Answering questions about itself

Self-questions ("Are you neuromorphic?", "Do you improve yourself?", "What is §21?") are matched by rule to an aspect or a section number and answered with the label and the sources: the section numbers and the file and symbol evidence. Answers are deterministic: the same question about the same program gives the same answer. A disclosure filter removes anything in an answer that resembles a key or credential, and the layer never reads settings or keys.

56.8 Interfaces

The layer is available in prototype v9.64 and in each application (Language, STEM, History and News, Control Panel) as GET /api/self (summary, label counts, aspects, process, sections, contradictions, gaps) and GET /api/self/ask?q=…&level=… (one answer with its sources); as the Self-knowledge page in v9.64 and the Self-knowledge tab of the Control Panel; and as a rule of the Control Panel assistant.

56.9 Status

The Self-Knowledge Layer is built and tested in prototype v9.64 (qb_selfknow.py, 1.0.0) and in the shared core of the applications (qbselfknow.py, 1.0.0). Its self-test (28 checks) runs inside every application's self-test; at release it reported no contradictions in either codebase.

Additions of October 8, 2026, highlighted in yellow: Sections 50–55 and the paragraphs headed "Implementation update", reflecting QueryBook prototype v9.64 and the UFCS-FQL laboratory prototype. No existing text was changed.

Additions of October 22, 2026, highlighted in green: Section 56, The Self-Knowledge Layer (SKL), built in prototype v9.64 (qb_selfknow.py) and the applications' shared core (qbselfknow.py); Composite Feature Registry v70 features 383–391; provisional FIG. 57 and claims 358–362. No existing text was changed.